

Role of Resistance Training Variables in Hypertrophy

A number of research-based methods exist for examining the muscular response to mechanical stimuli. For example, synergist ablation of the gastrocnemius muscle results in the soleus and plantaris muscles being forced to carry out plantar flexion. The heightened load on these muscles results in increases in muscle cross-sectional area of 30% to 50% within several weeks post-surgery. Neuromuscular electrical stimulation also is frequently used to promote hypertrophy in animal models. This technique, which involves stimulating muscles with high-frequency electrical impulses (levels above 60 Hz), produces significant gains in muscle mass in just a few sessions. In humans, however, resistance training is the primary means for increasing muscle growth.

Resistance training programs are a composite of *program design variables* that include volume, frequency, load, exercise selection, type of muscle action, rest interval length, repetition duration, exercise order, range of motion, and intensity of effort. These variables can be manipulated to stimulate the neuromuscular system, and they do so in different ways. Consistent with the *SAID principle* (specific adaptations to imposed demands), the way such stimuli are applied influences phenotypic adaptations. This chapter provides an overview of each variable with respect to how its manipulation affects the hypertrophic response to resistance training.

Note that to strengthen the ability to draw causal inferences, studies generally attempt to

manipulate a given variable while controlling all other variables. Although this is beneficial for research, in practice there is an interaction between variables, and manipulating one variable tends to affect the others. Thus, while each variable will be discussed in isolation, the implications of their manipulation must be taken into account within the context of the other variables when designing hypertrophy-oriented programs such as those discussed in chapter 8.

Volume

Volume refers to the amount of exercise performed over a period of time. Volume is often expressed as the number of repetitions completed in a resistance training bout (sets × repetitions). However, this value does not take into account the amount of load lifted. Thus, a more appropriate term to reflect the total work completed is *volume load*, which is the product of sets × repetitions × load. Although an increase in training frequency can create the largest increase in weekly volume load, provided volume per session is kept static, an increase in the number of sets performed (and thus total repetitions) in a training bout can also substantially increase training volume (99). Despite the relevance of volume load, resistance training volume for hypertrophy is most often expressed as *set volume*, operationally defined as the number of sets performed per muscle group over a given period of time, generally per week. This approach was reported to be viable

for quantifying resistance training volume when the repetition range lies between 6 and at least 20, training is carried out to failure, and all other variables are held constant (16).

Research provides compelling evidence that higher training volumes are necessary to maximize anabolism. This relationship has been demonstrated in multiple lines of evidence. For one, studies generally show heightened anabolic intracellular signaling studies with higher volumes. Terzis and colleagues (252) showed that phosphorylation of p70^{S6K} and ribosomal protein S6 increases 30 minutes following resistance training in a volume-dependent manner. The fact that values did not reach a plateau in the volumes studied suggests that higher volumes might have led to even greater increases. Intriguingly, the study found similar elevations in mTOR independent of training volume, suggesting that increased training volumes may augment S6 phosphorylation via alternative anabolic pathways, anticatabolic pathways, or a combination of the two. Consistent with these findings, Ahtiainen and colleagues (2) reported that markers of mTORC1 and p70^{S6K} increased to a greater extent after performing 10 sets versus 5 sets at 10RM. Similar results were recently shown when performing 6 versus 2 sets, with significantly greater phosphorylation of mTOR (12%), S6 kinase 1 (19%), and ribosomal protein S6 (28%) observed for the higher-volume condition (92).

Evidence also shows a volume-dependent effect on the muscle protein synthetic response to an acute training bout. This was demonstrated by Burd and colleagues (29), who found significantly greater increases in muscle protein synthesis 5 hours after 3 sets of knee extension exercises versus a single set (3.1- vs. 2.3-fold, respectively). Moreover, muscle protein synthesis in the 3-set condition remained significantly elevated (by 2.3-fold) at 29 hours post-workout, whereas levels in the 1-set condition had returned to baseline. In contrast to the aforementioned studies, however, phosphorylation of S6 was similar in the 1- and 3-set conditions. The combined findings from these studies indicate that multiple-set protocols in resistance training programs have greater positive effects

on intracellular signaling and muscle protein synthesis than single-set protocols.

Training volume affects the satellite cell response as well. Hanssen and colleagues (93) reported a greater increase in the number of satellite cells in the quadriceps femoris after 11 weeks of performing 18 sets compared to 6 sets of knee extension exercises per week. However, no significant differences were seen in the upper-body musculature despite similar volume differences, suggesting the hypertrophic influence of volume is more pronounced in the leg musculature. These findings are consistent with previous data from the same cohort showing significantly greater hypertrophy from a multiple- versus single-set protocol in the lower body (11% vs. 7%, respectively), whereas no significant differences were noted in the upper-body musculature (196). The studies were carried out with untrained subjects, so whether discrepancies persist in those with considerable lifting experience remains unclear.

The prevailing body of evidence from longitudinal studies parallels evidence from the acute study data. A systematic review by Wernbom and colleagues (267) carried out in 2007 showed that the cross-sectional area of the elbow flexors increased from 0.15% per day when 7 to 38 repetitions were performed per session to 0.26% per day when 42 to 66 repetitions were performed per session. The rate of increase diminished to 0.18% per day with volumes in the range of 74 to 120 repetitions per session, suggesting that very high volumes impair the hypertrophic response, perhaps by causing an overtrained state. With respect to total sets, hypertrophic increases peaked between 4 and 6 sets per session (0.24% increase in cross-sectional area per day); lesser responses were noted from the performance of 3 to 3.5 sets and ≥ 9 sets (0.17% and 0.18% increase per day, respectively). With respect to the quadriceps, the findings were similar across a wide spectrum of clusters; 0.12% to 0.13% increases in cross-sectional area per day were seen from the performance of 21 to 100+ repetitions per session. The only exception was in the cluster of 66 to 90 repetitions per day, in which cross-sectional area increases were on the

KEY POINT

Multiset protocols favoring high volumes of resistance training optimize the hypertrophic response. To avoid overtraining, people should increase volume progressively over the course of a training cycle and integrate periods of reduced training volume (i.e., deloads) regularly to facilitate the recovery process.

order of 0.08% per day. Analysis of the optimal number of sets showed a benefit to higher volumes, and the greatest response was seen in studies incorporating ≥ 10 sets per session. Importantly, the vast majority of these studies were carried out in untrained subjects, thereby limiting the ability to generalize findings to trained lifters.

More recently, a meta-analysis from our group (215) quantified the pooled data from 15 studies meeting inclusion criteria and found significantly greater hypertrophic increases when comparing higher to lower resistance training volumes. Stratification of volume into <5 , 5 to 9, and 10+ sets per week showed a dose-response relationship. Higher volumes correlated to greater increases in muscle mass; the graded increases were noted in percentage gains across categories (5.4%, 6.6%, and 9.8%, respectively). Moreover, subgroup analysis showed that results strengthened with the use of more accurate site-specific methods of hypertrophy measurement (e.g., MRI, ultrasound). The totality of findings provides compelling evidence that volume is a primary driver of muscle hypertrophy. However, a lack of data investigating the effects of higher training volumes precludes the ability to determine whether additional hypertrophic gains could be achieved from >10 weekly sets per muscle and, if so, at what point it would reach a threshold.

Table 4.1 summarizes the research related to volume and muscle hypertrophy.

Although the evidence for a dose-response relationship is compelling, there is undoubtedly a limit above which additional volume confers no additional hypertrophic benefits. A

number of bodily systems, including metabolic, hormonal, nervous, and muscular, are sensitive to the magnitude of training volume (125), and overstressing these systems is bound to have negative consequences on adaptations. The relationship between volume and hypertrophy is hypothesized to follow an inverted-U curve, whereby muscle accretion peaks at a given volume load and, beyond this point, further increases in volume can actually impair muscular gains (figure 4.1) (99). It is important to note that the threshold for volume-related hypertrophic benefits varies based on genetics (see chapter 7); lifestyle-related factors such as nutritional status, daily stress levels, and sleep patterns also play a role in individual responses. Some authors have posited that well-trained lifters require a particularly high training volume (>10 sets) to induce maximal hypertrophy (176), although this hypothesis remains controversial.

Since publication of our meta-analysis (215), which included studies published up to December 2014, several additional studies have investigated the limits of a volume threshold in people with previous resistance training experience. Some of these studies indicate that the volume threshold may extend up to 30+ per muscle per week (95, 186, 218), whereas others show a plateau at 10 or fewer sets (14). While it is difficult to reconcile the discrepancies

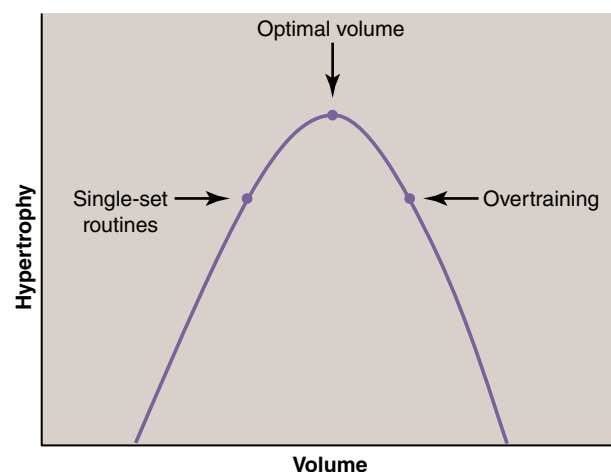


FIGURE 4.1 Dose response for the effects of volume on hypertrophy.

TABLE 4.1 Summary of Hypertrophy Training Studies Investigating Training Volume

Study	Subjects	Design	Study duration	Hypertrophy measurement	Findings
Amirthalingam et al. (6)	19 trained young men	Random assignment to a resistance training protocol in which one group performed 5 or 10 sets of various compound exercises for the major muscle groups performed in a split-routine. Subjects also performed 3 to 4 sets of additional targeted exercises for each muscle group. Training was carried out at 10RM for each exercise 3 days per week.	6 weeks	Ultrasound (elbow flexors, elbow extensors, quadriceps, hamstrings), DXA	No significant differences in muscle thickness between conditions. Increases in trunk and arm lean soft-tissue mass favored the lower volume condition.
Barbalho et al. (13)	37 trained young men	Random assignment to a resistance training protocol comprising either 5, 10, 15, or 20 sets per muscle per week. All subjects performed a total-body training program carried out at 10RM for each exercise 3 days per week.	24 weeks	Ultrasound (biceps brachii, triceps brachii, pectoralis major, quadriceps, gluteus maximus)	No significant differences in muscle volume between conditions. Relative changes were greater in the lower-volume conditions vs. the higher-volume conditions.
Barbalho et al. (14)	40 trained young women	Random assignment to a resistance training protocol comprising either 5, 10, 15, or 20 sets per muscle per week. All subjects performed a total-body training program carried out at 10RM of each exercise 3 days per week.	24 weeks	Ultrasound (biceps brachii, triceps brachii, pectoralis major, quadriceps, gluteus maximus)	No significant differences in muscle volume between conditions. Relative changes were greater in the lower-volume conditions vs. the higher-volume conditions.
Bottaro et al. (26)	30 untrained young men	Random assignment to a resistance training protocol in which one group performed 3 sets of knee extension exercises and 1 set of elbow flexion exercises, while the other group performed 3 sets of elbow flexion exercises and 1 set of knee extension exercises. All subjects performed 8RM to 12RM of each exercise twice per week.	12 weeks	Ultrasound (elbow flexors, quadriceps)	No significant differences in muscle thickness between conditions
Cannon and Marino (41)	31 untrained young and elderly women	Random assignment to a resistance training protocol of either 1 or 3 sets per exercise. Exercise consisted of bilateral knee extensions and knee curls for 10 reps at an intensity of 50% to 75% of 1RM. Training was carried out 3 days per week.	10 weeks	MRI (quadriceps)	No significant differences in muscle volume between conditions
Correa et al. (51)	35 untrained postmenopausal women	Random assignment to a resistance training protocol of either 1 or 3 sets per exercise. All subjects performed 8 exercises targeting the entire body at 15RM.	12 weeks	Ultrasound (quadriceps)	No significant differences in quadriceps muscle thickness between conditions. Relative increases in muscle thickness favored the higher-volume condition for the vastus medialis, vastus lateralis, and rectus femoris.
Galvao et al. (75)	28 untrained elderly men and women	Random assignment to a resistance training protocol of either 1 or 3 sets per exercise. All subjects performed 7 exercises targeting the entire body at 8RM. Training was carried out twice weekly.	20 weeks	DXA	No significant differences in lean body mass between conditions

(continued)

Table 4.1 (continued)

Study	Subjects	Design	Study duration	Hypertrophy measurement	Findings
Hackett et al. (88)	12 trained young men	Random assignment to a resistance training protocol in which one group performed 5 or 10 sets of various compound exercises for the major muscle groups performed in a split routine. Subjects also performed 3 to 4 sets of additional targeted exercises for each muscle group. Training was carried out at 10RM for each exercise 3 days per week.	12 weeks	DXA	No significant differences in lean body mass between conditions. Increases in lean leg mass favored the higher-volume condition for the first 6 weeks, but these gains were wiped out during the final 6 weeks of the program.
Heaselgrave et al. (97)	49 trained young men	Random assignment to a resistance training protocol comprising either 9, 18, or 27 sets of biceps exercise per week. Subjects performed 10 to 12 reps per set with training carried out 1 or 2 days per week.	6 weeks	Ultrasound (elbow flexors)	No significant differences in muscle thickness between conditions. Greater relative increases in muscle thickness were observed in the 18-set condition.
Marzolini et al. (141)	53 untrained elderly men and women with coronary artery disease	Random assignment to a resistance training protocol of either 1 or 3 sets per exercise. All subjects performed 10 exercises targeting the entire body for 10 to 15 reps.	24 weeks	DXA	Markedly greater increases in lean body mass, lean arm mass, and lean leg mass for the high-volume condition
McBride et al. (146)	28 untrained young men and women	Random assignment to a resistance training protocol consisting of 5 exercises, including the biceps curl, leg press, chest fly, sit-up, and back extension. One group performed a single set of each of these exercises, while the other group performed 6 sets for the biceps curl and leg press and 3 sets for all other exercises. All subjects performed 6RM to 15RM of the exercises twice weekly.	12 weeks	DXA	No significant differences in lean body mass between conditions
Mitchell et al. (153)	18 untrained young men	Random assignment to perform 2 of 3 unilateral knee extension protocols: 3 sets at 30% of 1RM, 3 sets at 80% of 1RM, 1 set at 80% of 1RM. Each participant trained both legs and was thus assigned to 2 of the 3 possible training conditions. Training was carried out 3 days per week.	10 weeks	MRI (quadriceps)	No statistically significant differences in quadriceps hypertrophy between conditions, although the high-volume condition experienced more than double the relative growth of the low-volume condition.
Munn et al. (160)	115 untrained young men and women	Random assignment to a resistance training protocol of 1 or 3 sets per exercise in either a slow or fast fashion. Training was carried out using elbow flexion exercises at 6RM to 8RM for 3 days per week.	6 weeks	Skinfold and circumference measurements	No significant differences in lean mass between conditions
Ostrowski et al. (174)	27 resistance-trained young men	Random assignment to a resistance training protocol of 1, 2, or 4 sets per exercise. All subjects performed a 4-day split-body routine, working each of the major muscle groups with multiple exercises in a session at 7RM to 12RM.	10 weeks	Ultrasound (elbow extensors, quadriceps)	No significant differences in muscle thickness between conditions. Greater relative increases in quadriceps muscle thickness favoring the highest volume condition.

Study	Subjects	Design	Study duration	Hypertrophy measurement	Findings
Radaelli et al. (184)	20 untrained elderly women	Random assignment to a resistance training protocol of either 1 or 3 sets per exercise. All subjects performed 10 exercises targeting the entire body at 10RM to 20RM. Training was carried out twice weekly.	13 weeks	Ultrasound (elbow flexors, quadriceps)	No significant differences in muscle thickness between conditions
Radaelli et al. (186)	48 recreationally trained young men	Random assignment to a resistance training protocol of 1, 3, or 5 sets per exercise. All subjects performed 8RM to 12RM for multiple exercises for the entire body. Training was carried out 3 days per week.	6 months	Ultrasound (elbow flexors, elbow extensors)	Significantly greater increases in elbow flexor muscle thickness for the 5-set condition compared to the other two conditions. Only the 3- and 5-set groups significantly increased elbow flexor muscle thickness from baseline. Significantly greater increases in elbow extensor muscle thickness in the 5-set condition compared with the other two conditions. Only the 5-set group significantly increased elbow extensor thickness from baseline.
Radaelli et al. (187)	27 untrained elderly women	Random assignment to a resistance training protocol of either 1 or 3 sets per exercise. All subjects performed 10 exercises targeting the entire body at 10RM to 20RM. Training was carried out twice weekly.	6 weeks	Ultrasound (quadriceps)	No significant differences in muscle thickness between conditions
Radaelli et al. (185)	20 untrained elderly women	Random assignment to a resistance training protocol of either 1 or 3 sets per exercise. All subjects performed 10 exercises targeting the entire body at 6RM to 20RM. Training was carried out twice weekly.	20 weeks	Ultrasound (elbow flexors, quadriceps)	Significantly greater increases in quadriceps thickness for the high-volume group
Rhea et al. (191)	18 resistance-trained young men	Random assignment to a resistance training protocol of either 1 or 3 sets per exercise. All subjects performed 4RM to 10RM on the bench press and leg press. Subjects also performed an additional set of multiple exercises considered unrelated to the bench press or leg press. Training was carried out 3 days per week.	12 weeks	BodPod	No significant differences in lean body mass between conditions
Ribeiro et al. (192)	30 untrained elderly women	Random assignment to a resistance training protocol consisting of 8 exercises for the total body performed either 1 or 3 days per week. All subjects performed 10 to 15 reps.	12 weeks	DXA	No significant differences in lean body mass between conditions
Rønnestad et al. (196)	21 untrained young men	Random assignment to a resistance training protocol in which one group performed 3 sets of upper-body exercises and 1 set of lower-body exercises, while the other group performed 3 sets of lower-body exercises and 1 set of upper-body exercises. Training consisted of 8 exercises for the entire body performed at 7RM to 10RM and carried out 3 days per week.	11 weeks	MRI (trapezius, quadriceps)	Significantly greater increases in thigh muscle cross-sectional area for the higher-volume condition

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Table 4.1 (continued)

Study	Subjects	Design	Study duration	Hypertrophy measurement	Findings
Schoenfeld et al. (218)	34 trained young men	Random assignment to a total body resistance training protocol performed with either 1, 3, or 5 sets per exercise. All subjects performed 8 to 12 reps for 3 days per week.	8 weeks	Ultrasound (elbow flexors, elbow extensors, quadriceps)	Significant increases favoring the higher-volume conditions were seen for the elbow flexors, midthigh, and lateral thigh.
Sooneste et al. (235)	8 untrained young men	Within-subject crossover in which all subjects performed a 2-day-per-week resistance training protocol of preacher curls so that one arm used 3 sets in a session and the other arm used a single set in the following session. Training was performed at 80% of 1RM and carried out 2 days per week.	12 weeks	MRI (elbow flexors)	Significantly greater increases in upper-arm cross-sectional area for the high-volume condition
Starkey et al. (239)	48 untrained mixed-aged men and women	Random assignment to a resistance training protocol of knee flexions and extensions performed either 1 or 3 days per week. All subjects performed 8 to 12 reps for 3 days per week.	14 weeks	Ultrasound (quadriceps, hamstrings)	No significant differences in the thickness of the anterior or posterior thigh muscles between conditions, although only the high-volume group significantly increased hypertrophy of the vastus medialis relative to control.

Abbreviations: RM = repetition maximum; DXA = dual X-ray absorptiometry; MRI = magnetic resonance imaging.

PRACTICAL APPLICATIONS

VOLUME

Evidence for a dose–response relationship between volume and hypertrophy is compelling: Higher training volumes are positively associated with greater muscular gains. A volume of approximately 10 to 20 sets per muscle per week appears to be a good general recommendation for hypertrophy-related goals. More advanced lifters seem to require greater volumes to maximize muscle protein accretion and thus might need to train at the higher end of these recommendations; experimentation is warranted to determine individual responsiveness. There may be a benefit to selectively employing even higher volumes to bring up lagging muscle groups. Given that consistently employing high volumes over time hastens the onset of overtraining, periodizing programming by progressively increasing volume over the course of a training cycle appears beneficial. Moreover, periods of reduced training volume should be integrated regularly to facilitate the recovery process and resensitize muscle tissue.

between studies, a possible explanation may be related to the composition of the routines in the studies. Specifically, studies showing beneficial effects for very high volumes employed total-body workouts in which the volume for each muscle was spread out over the course of the week. Alternatively, studies showing no

additional benefits to higher-volume training used split routines with the volume for each muscle condensed into a single workout. An unpublished simulation analysis suggests that a threshold for per-session volume exists at approximately 10 sets per week; beyond this point, additional volume appears to confer

minimal further hypertrophic benefits (personal communication). This hypothesis warrants further study.

It is important to note that studies investigating volume are generally specific to given muscle groups, and results therefore cannot be generalized to all muscle groups for training programs as a whole. For example, in a study from my lab (218) showing hypertrophic benefits from 30+ sets per muscle per week, the total training time for the highest-volume group was just ~3.5 hours per week. The apparent paradox can be explained by the fact that the study specifically focused on assessing hypertrophy of the muscles of the arms and legs, and hence included only 7 exercises per session. So while the muscles of interest received high volumes, others were worked at much more modest volumes.

The response to different volumes is highly individual. In perhaps the most elegant study on the topic to date, Hammarström and colleagues (92) employed a within-subject design in which untrained subjects were randomized to perform a higher volume with one leg (~15 sets per muscle per week) and a lower volume with the other leg (~5 sets per muscle per week). After 12 weeks of training, the higher-volume

condition elicited significantly greater quadriceps hypertrophy than the lower-volume condition. Moreover, these changes coincided with a greater activation of anabolic intracellular signaling pathways and a heightened stimulation of ribosome biogenesis. Most interestingly, ~44% of the cohort derived a clear benefit from the higher-volume condition while only ~9% showed a clear benefit from the lower-volume approach; the remaining subjects (~47%) showed similar responses irrespective of training volume. These findings are especially relevant given the within-subject design whereby subjects served as their own controls, thus reducing the potential confounding influence from individual variability. Consistent with these results, a study from my lab (218) also showed fewer poor responders when training at higher volumes than when training at lower volumes based on analysis of the smallest worthwhile change (figure 4.2).

When considering the body of literature as a whole, as well as taking practical considerations into account, a volume of approximately 10 to 20 sets per muscle per week appears to be a good general recommendation to maximize hypertrophy. Some may thrive with

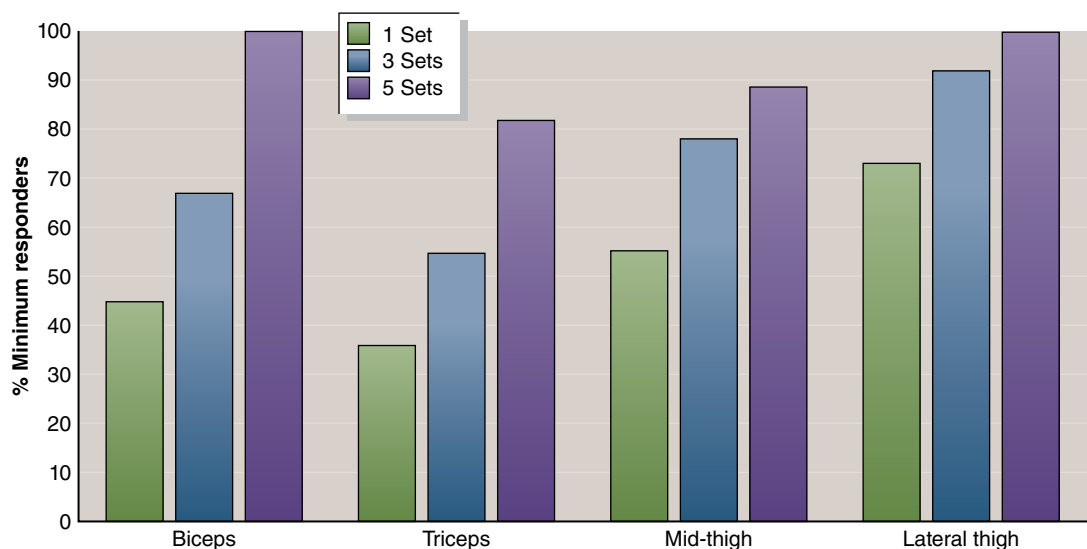


FIGURE 4.2 Percent responders in 1 vs. 3 vs. 5 sets per exercise.

Data from Schoenfeld et al. (218).

slightly lower volumes and others thrive with somewhat higher volumes; experimentation is warranted to determine individual responsiveness. Given data indicating that responsiveness to higher volumes shows a dose–response relationship, there may be a benefit to strategically employing higher volumes for a poorly responding muscle group. For example, if the muscle development of the deltoids is lagging behind other groups, increasing their volume of training above that of other muscle groups could be warranted. If multiple muscles are considered poor responders, employ higher volumes for a single muscle group at a time over a given training cycle; set up a rotation that targets the other lagging muscle groups with higher volumes in future cycles.

Frequency

Frequency of training pertains to the number of exercise sessions performed in a given period of time, generally a week (205). Perhaps more important to hypertrophic outcomes, frequency also includes the number of times a muscle group is worked over the course of a week. With respect to hypertrophy training, frequency can be varied to manipulate training volume. Neuromuscular factors limit how much volume can be incorporated into a single training session; beyond a given threshold, the quality of training begins to degrade. Studies show superior neuromuscular adaptations, hormonal markers for recovery, strength improvement, and gains in lean body mass in those performing volume-equated programs with higher frequencies and less volume per session (99). Thus, distributing volume per muscle group over more frequent bouts can be an effective strategy for maintaining weekly volume with less fatigue per session.

Hypertrophy-oriented routines generally involve a high volume of work per muscle group in a session but relatively infrequent training of each muscle group. To best carry out this strategy, people often follow a *split-body routine* in which they perform multiple

exercises for a specific muscle group in one training session. In comparison to a total-body routine, split routines allow total weekly training volume to be maintained or increased with fewer sets performed per training session and greater recovery afforded between sessions (119). Moreover, performing multiple exercises for a muscle group in the same bout heightens metabolic stress and thus may enhance anabolism (205). A survey of competitive male bodybuilders revealed that more than 2/3 trained each muscle group only once per week, and none reported working a muscle group more than twice weekly; every respondent reported using a split-body routine (87).

General hypertrophy training guidelines recommend allowing at least 48 hours between resistance bouts for the same muscle group (205). It has been surmised that training before muscle protein synthesis has fully run its course—which lasts up to ~48 hours post-exercise—impairs muscle protein accretion (135). Research in rodents shows that myogenic responses are attenuated when recovery occurs less than 48 hours after the previous resistance bout (89). Moreover, total RNA has been shown to be elevated in humans 72 hours after a bout of maximal isometric electrical contractions (21). Because the majority of skeletal muscle RNA is ribosomal, these findings suggest that a cell's potential for protein synthesis remains heightened even beyond the 2-day time point.

The extent of perturbations to exercised muscle also mitigates training frequency. Metabolically fatigued muscle fibers display a greater membrane permeability consequent to an increase in free calcium ions, leading to the activation of potassium channels and proteolytic enzymes. Performing a multiset, high-volume routine consistent with hypertrophy training protocols may thus require at least 48 to 72 hours of rest between workouts for the same muscle group to ensure adequate repair, recovery, and adaptation (126, 136). However, these findings do not take into account the adaptive capacity of the neuromuscular system, whereby protective mechanisms (i.e.,

KEY POINT

Split routines allow for a greater volume of work per muscle group per session, potentially enhancing muscular adaptations via the dose–response relationship between volume and hypertrophy.

the repeated bout effect) ameliorate ultrastructural myodamage.

A 2007 systematic review by Wernbom and colleagues (267) determined that although novice lifters benefit from training muscle groups up to 4 days a week, those with more experience realize optimal gains with a weekly frequency of 2 or 3 days. There was insufficient data for determining whether higher frequencies would be beneficial in a well-trained population. However, these findings were based on limited data. Importantly, the analysis did not account for greater volumes associated with higher training frequencies, thereby confounding the ability to draw conclusions on the specific impact of varying the number of weekly training sessions.

Since publication of the Wernbom and colleagues review (267), an emerging body of research has been published examining the effects of frequency on long-term hypertrophic adaptations in humans. Our group performed a meta-analysis of the current data, which at the time comprised 25 studies that directly compared higher versus lower resistance training frequencies (219). When volume was equated between conditions, results showed similar hypertrophic changes regardless of whether muscle groups were worked 1, 2, 3, or 4+ days per week. Alternatively, pooling data from studies whereby volume was not equated showed a small but significant benefit for higher training frequencies (although there was insufficient data to evaluate whether effects persisted for frequencies above 3 days per week). These findings indicate that, as a standalone variable, frequency does not have much impact on muscle development; it seems that its primary utility is to act as a vehicle to manage weekly volume.

The value of spreading volume across greater weekly frequencies appears to become increasingly important with the implementation of higher training volumes. As mentioned in the previous section on volume, evidence points to a per-session volume threshold of approximately 10 sets per muscle group, with diminishing value in performing additional sets. Thus, when performing a target volume of, say, 20 sets per muscle per week, greater muscular adaptations are attained by apportioning volume into two weekly sessions of 10 sets for a given muscle group as opposed to a single session of 20 sets. The implementation of higher volumes for a given muscle group (e.g., 30 sets) would theoretically necessitate even higher weekly frequencies for training that muscle (e.g., 3 days per week).

A popular strategy to increase volume by manipulating training frequency is to split up a workout by performing multiple sessions in a day (often morning and evening). This strategy, called a *double-split routine*, is commonly used by bodybuilders to allow for high weekly training volumes while maintaining optimal mental and physical abilities during training. A study by Häkkinen and Kallinen (90) lends support to the value of double splits for hypertrophy training. Employing a crossover design, female athletes performed 2 training blocks lasting 3 weeks each. The athletes trained once a day during the first block and twice a day during the second block. The training volume was the same for each block, and training occurred 3 days per week. Results showed greater increases in muscle cross-sectional area when the athletes performed 2 sessions per day rather than when they performed all sets in a single bout. Conversely, Hartman and colleagues (94) found that once-daily training produced slightly greater cross-sectional area increases compared to twice-daily splits in a group of nationally competitive male weightlifters over a 3-week period, although the differences were not statistically significant. Both of these studies were of very short duration, limiting the ability to draw practical conclusions on the topic. The conflicting results leave open the possibility that double-split routines are a viable option for hypertrophy training provided that the person can fit such an approach into his or her daily schedule.

Some researchers have speculated that very frequent training sessions comprised of low per-session volumes may help to maximize the hypertrophic response. This hypothesis is based on the premise that the muscle protein synthetic response to a training bout is truncated as an individual gains training experience (55). Indeed, while muscle protein synthesis in untrained individuals remains elevated for ≥ 48 hours (177), trained lifters experience a higher initial peak response that returns to baseline after < 28 hours (243). Based on these data, more repetitive stimulation of the muscle with higher weekly training frequencies would seem to allow for greater time spent in a net-positive protein balance, conceivably promoting a greater accretion of muscle proteins over time.

However, the aforementioned reports of a blunted anabolic response in trained individuals are specific to mixed measures of muscle protein synthesis, which comprise all myocellular proteins, not just contractile elements. Research indicates that whereas post-exercise synthesis of noncontractile proteins is diminished with resistance training experience, the increases in synthesis of myofibrillar proteins are largely preserved (120). Accordingly, studies show that measures of myofibrillar protein synthesis remain elevated in trained lifters for at least 48 hours, if not longer (54). This calls into question the rationale on which the high training frequency hypothesis is based.

Proponents of very high frequency training often point to a study carried out on Norwegian powerlifters (the Norwegian Frequency Project) as evidence supporting the strategy (183). The study remains unpublished and has been presented only as a conference abstract. Based on what can be gleaned from available information, members of the Norwegian

national powerlifting team were randomized to perform the squat, deadlift, and bench press either for 4 sets across 3 nonconsecutive days per week or 2 sets across 6 consecutive days per week. After 15 weeks, the higher-frequency condition increased in quadriceps cross-sectional area by 4.2%, whereas the lower-frequency condition showed a slight decrease (-0.6%), despite an equated total volume between conditions. While the findings seem to support a hypertrophic benefit to more frequent training stimuli, it should be noted that sets were stopped well short of muscle failure, and thus the protocol does not reflect training practices consistent with hypertrophy-oriented routines.

A recent study endeavored to replicate the essence of the Norwegian Frequency Project, but the design employed a protocol more common to bodybuilding programs (201). As with the aforementioned study (183), subjects were randomized to perform 4 sets of each exercise across 3 nonconsecutive days per week or 2 sets across 6 consecutive days per week. However, a variety of multi- and single-joint exercises were included in the protocol, and sets were carried out until volitional failure. Results showed similar increases in muscle thickness of the triceps and quadriceps between conditions. Alternatively, elbow flexor growth in the lower-frequency condition was significantly greater than that of the higher-frequency condition, which in fact did not show any change in muscle thickness over the 8-week study period. Thus, support for the use of very high frequency training routines to enhance muscle development remains equivocal.

Table 4.2 provides a summary of the research related to training frequency and muscle hypertrophy.

TABLE 4.2 Summary of Hypertrophy Training Studies Investigating Training Frequency

Study	Subjects	Frequency	Study duration	Volume equated?	Hypertrophy measurement	Findings
Arazi and Asadi (8)	39 untrained young men	1 vs. 2 vs. 3 days per week	8 weeks	Yes	Circumference measurements	No significant differences in arm or thigh girth between conditions
Barcelos et al. (15)	20 trained young men	2 vs. 3 vs. 5 days per week	8 weeks	No	Ultrasound (vastus lateralis)	No significant differences in muscle thickness between groups
Benton et al. (20)	21 untrained middle-aged women	3 vs. 4 days per week	8 weeks	Yes	BodPod	No significant differences in lean body mass between conditions
Brigatto et al. (27)	20 untrained young men	1 vs. 2 days per week	8 weeks	Yes	Ultrasound (elbow flexors, elbow extensors, vastus lateralis, anterior quadriceps)	No significant differences in muscle thickness between groups
Calder et al. (38)	30 untrained young women	2 vs. 4 days per week	20 weeks	Yes	DXA	No significant differences in lean mass between groups
Candow and Burke (40)	29 untrained middle-aged men and women	2 vs. 3 days per week	6 weeks	Yes	DXA	No significant differences in lean body mass between conditions
Carneiro et al. (44)	53 untrained elderly women	2 vs. 3 days per week	12 weeks	No	DXA	No significant differences in skeletal muscle mass between groups
Colquhoun et al. (50)	28 trained young men	3 vs. 6 days per week	6 weeks	Yes	Ultrasound (FFM)	No significant differences in FFM between conditions. Relative increases in FFM favored the higher-frequency condition.
Fernandez-Lezaun et al. (68)	68 untrained older men and women	1 vs. 2 vs. 3 days per week	24 weeks	No	DXA	No significant differences in FFM between conditions
Gentil et al. (76)	30 untrained young men	1 vs. 2 days per week	10 weeks	Yes	Ultrasound (elbow flexors)	No significant differences in elbow flexor thickness between groups
Gentil et al. (77)	16 trained young men	1 vs. 2 days per week	10 weeks	Yes	Ultrasound (elbow flexors)	No significant differences in muscle thickness between groups. Only the 1-day-per-week condition significantly increased muscle thickness from baseline.
Gomes et al. (81)	23 trained young men	1 vs. 5 days per week	8 weeks	Yes	DXA	No significant differences in FFM between conditions
Häkkinen et al. (90)	10 resistance-trained young women	Once daily vs. twice daily	3 weeks	Yes	Ultrasound (quadriceps)	Greater increases in quadriceps thickness in the higher-frequency condition
Hartman et al. (94)	10 elite-level male weightlifters	Once daily vs. twice daily	3 weeks	Yes	Ultrasound (quadriceps)	No significant differences in thickness of the quadriceps femoris between conditions

(continued)

Table 4.2 (continued)

Study	Subjects	Frequency	Study duration	Volume equated?	Hypertrophy measurement	Findings
Lasevicius et al. (130)	36 trained young men	2 vs. 3 days per week	10 weeks	Yes	Ultrasound (elbow flexors, elbow extensors, quadriceps)	No significant differences in thickness between conditions. Effect size differences favored the lower-frequency condition for all measures.
McLester et al. (149)	25 recreationally trained young men and women	1 vs. 3 days per week	12 weeks	Yes	Skinfold technique, circumference measurements	Nonsignificant trend for greater increases in lean mass in the higher-frequency condition
Murlasits et al. (161)	24 untrained older men and women	2 vs. 3 days per week	8 weeks	No	DXA	No significant differences in FFM between conditions
Nascimento et al. (164)	45 untrained older women	2 vs. 3 days per week	12 weeks	No	DXA	No significant differences in skeletal muscle mass between conditions
Ochi et al. (171)	20 untrained young men	1 vs. 3 days per week	11 weeks	Yes	Ultrasound (vastus lateralis, rectus femoris, vastus medialis, vastus intermedius)	No significant differences in quadriceps muscle thickness between conditions
Pina et al. (179)	39 untrained older women	2 vs. 3 days per week	24 weeks	No	DXA	No significant differences in lean mass between conditions. Relative differences favored the 2-days-per-week condition.
Ribeiro et al. (193)	10 elite male bodybuilders	2 vs. 3 days per week	4 weeks	Yes	DXA	No significant differences in lean mass between conditions
Richardson et al. (194)	40 untrained older men and women	1 vs. 2 days per week performed with either high velocity/low loads or low velocity/high loads	10 weeks	No	BIA	Significantly greater increases in FFM 1 day per week with low velocity/high load compared to 2 days per week using high velocity/low load. Overall, increases in FFM were minimal across all groups.
Saric et al. (201)	27 trained young men	3 vs. 6 days per week	6 weeks	Yes	Ultrasound (elbow flexors, elbow extensors, rectus femoris, vastus intermedius)	No significant difference in elbow extensor or quadriceps muscle thickness between conditions. Elbow flexor muscle thickness was significantly greater in the 3-days-per-week condition.
Schoenfeld et al. (212)	19 resistance-trained young men	1 vs. 3 days per week	8 weeks	Yes	Ultrasound (elbow flexors, elbow extensors, quadriceps)	Significantly greater increases in elbow flexor muscle thickness and a trend for greater increases in vastus lateralis thickness for the greater-frequency, total-body condition

Study	Subjects	Frequency	Study duration	Volume equated?	Hypertrophy measurement	Findings
Serra et al. (226)	74 untrained young men and women	2 vs. 3 vs. 4 days per week	12 weeks	No	Skinfold measurements	No significant difference in lean mass between conditions
Taaffe et al. (241)	46 untrained older men and women	1 vs. 2 days per week	24 weeks	No	DXA	No significant difference in lean soft-tissue mass between conditions
Tavares et al. (246)	33 untrained young men	1 vs. 2 days per week	8 weeks	Yes	MRI (quadriceps)	No significant differences in quadriceps CSA between conditions
Thomas and Burns (254)	19 trained men and women	1 vs. 3 days per week	8 weeks	Yes	DXA	No significant differences in lean soft-tissue mass between conditions
Turpela et al. (256)	106 untrained older men and women	1 vs. 2 vs. 3 days per week	24 weeks	No	DXA, ultrasound (quadriceps)	No significant differences in lean soft-tissue mass or CSA between conditions
Yue et al. (275)	18 recreationally trained young men	1 vs. 2 days per week for lower body; 2 vs. 4 days per week for upper body	6 weeks	Yes	BodPod, ultrasound (elbow flexors, vastus medialis, anterior deltoids)	No significant difference in FFM or quadriceps muscle thickness. Only the lower-frequency condition significantly increased elbow flexor muscle thickness.
Zaroni et al. (276)	18 trained young men	1 vs. 5 days per week for lower body; 2 vs. 5 days per week for upper body	8 weeks	Yes	Ultrasound (elbow flexors, elbow extensors, vastus lateralis)	Greater increases in muscle thickness for all muscles studied in the 5-days-per-week condition

Abbreviations: RM = repetition maximum; DXA = dual X-ray absorptiometry; FFM = fat-free mass; CSA = cross-sectional area; MRI = magnetic resonance imaging.

PRACTICAL APPLICATIONS

FREQUENCY

Modulating training frequency is an effective way to manipulate training volume. When performing relatively low weekly volumes, frequency of training does not play much if any role in muscle growth, and individuals can choose the frequency that best fits their schedule and goals. Alternatively, when moderate to high volumes are performed (>10 sets per muscle per week), higher training frequencies (at least twice per week) provide better volume management and thus facilitate greater muscular adaptations. If very high volumes are implemented for a given muscle group (~30 sets per muscle per week), spreading training across at least 3 weekly sessions appears to be warranted. Very high training frequencies (6 days per week) do not appear to be more effective than moderately high frequencies (3 days per week) for enhancing hypertrophy, although limited evidence precludes the ability to draw strong inferences on the topic. Although both total-body and split routines can be viable training strategies, dividing workouts by body region (e.g., upper and lower, pushing and pulling) may be more effective when training with higher volumes because it allows higher weekly frequencies (and thus shorter sessions) while affording greater muscular recuperation between workouts.

Load

The load lifted is widely considered one of the most important factors in the hypertrophic response to resistance training. *Intensity of load* refers to the percentage of 1RM employed in a given exercise. For example, if someone has a maximal bench press of 100 lb (45.5 kg) and performs a set with 80 lb (36.4 kg), then the intensity of load would be expressed as 80% of 1RM.

Intensity of load is often categorized into loading zones that correspond to repetition ranges. Typically, repetition ranges are classified as heavy (1RM to 5RM), medium (6RM to 12RM), and light (15+RM) (205). Although formulas have been devised to estimate repetitions at a given percentage of 1RM, at best they provide only a crude approximation of the relationship between repetitions and the percentage of 1RM. The combination of genetic factors (e.g., muscle fiber typing, internal moment arm length), physiological factors (e.g., buffering capacity), and exercise types (e.g., upper body versus lower body, single joint versus multi-joint) affect the generalizability of values. Hoeger and colleagues (104) found that a load of 80% of 1RM corresponded to 10RM in the bench press, lat pulldown, and knee extension; however, this intensity of load varied from 6RM for the leg curl and 7RM to 8RM for the arm curl, to 15RM for the leg press. Moreover, the accuracy of these formulae declines substantially as loads become progressively lighter. To this end, another study showed that, for individual subjects, repetitions to failure in the leg press ranged between 7 and 24 at 75% of 1RM, whereas the disparity widened to 30 to 71 at 30% of 1RM (207).

In a 2007 systematic review, Wernbom and colleagues (267) concluded that maximal hypertrophy is achieved through the use of a medium-repetition range, a claim that has been echoed by other researchers (125, 205). This hypothesis is primarily based on an extrapolation of mechanistic factors associated with the hypertrophic response to resistance training. At the time of the review, only a limited number of studies had directly compared training with higher loading schemes to training with lower loading schemes.

Heavy loading is generally believed to promote neural adaptations and to have lesser

effects on hypertrophy (109). High intensities of load (>85% of 1RM) naturally result in high levels of mechanical tension on muscles. However, because the duration of a heavy set is short (<15 seconds), energy during such training is primarily derived from the ATP-PC system and little contribution occurs from fast glycolysis. Thus, metabolite accumulation is relatively low, which is supported by research showing that peripheral fatigue induced via metabolic stress was significantly reduced when training in a low-repetition range (5 repetitions per set) compared to sets carried out in a medium-repetition range (10 repetitions per set) (195).

At the other end of the loading zone continuum, light-load training is associated with high amounts of metabolic stress. Sets of ≥ 15 repetitions generally last 45 seconds or more, requiring the majority of energy production to be derived from the fast-glycolytic system. This results in a substantial buildup of metabolites and acidosis and generates a significant muscle pump. However, it has been theorized that the forces required to lift light loads are insufficient to recruit the highest-threshold motor units (207), which would mitigate hypertrophic gains.

Training in a medium-repetition range is purported to provide an optimal combination of mechanical tension and metabolic stress for maximizing hypertrophic adaptations. Loads during such training are heavy enough to recruit the majority of fibers in the target musculature and to maintain their stimulation over a sufficient period of time. Moreover, sets generally last between 20 and 40 seconds, requiring a substantial contribution from fast glycolysis and correspondingly generating higher levels of metabolic stress (58). Because of these factors, medium loading is often referred to as the *hypertrophy range*.

Despite having a sound logical basis, the concept of an optimal hypertrophy range has come under scrutiny. With respect to muscle recruitment, fast-twitch fibers begin to be activated when force exceeds 20% of maximal voluntary isometric contraction, and activation of the full motor unit pool occurs at approximately 80% of maximal voluntary isometric contraction (80, 253). Recruitment during traditional resistance training under dynamic conditions is less clear. Research shows corresponding increases in electromyography (EMG) amplitude during

KEY POINT

Training across a wide spectrum of repetition ranges (1 to 20+) is recommended to maximize all possible avenues for the complete development of the whole muscle. However, there is some merit in focusing on a medium-repetition range (6RM to 12RM), which may provide an optimal combination of mechanical tension and metabolic stress.

fatiguing contractions, ostensibly as a result of an increased contribution of higher-threshold motor units recruited to maintain force output (237). It has therefore been postulated that training to the point of concentric muscular failure, regardless of the magnitude of load, ultimately results in the recruitment of the full spectrum of available motor units (33, 45). However, although acknowledging that motor unit activity does increase with fatigue, others claim that lifting very heavy loads results in specific recruitment patterns that are not attainable with light-load training (125).

Research from surface EMG studies consistently show a reduced EMG amplitude in lower-load training than in higher-load training when carried out to failure (112, 158, 203, 207), with evidence indicating that a minimum load of approximately 70% of 1RM is required to achieve equated amplitudes (82). However, surface EMG represents the neural drive to a given muscle, which comprises not only motor unit recruitment but other factors such as rate coding, synchronization, muscle fiber propagation velocity, and intracellular action potentials as well (17, 57). In an effort to achieve better clarity on the topic, Muddle and colleagues (159) employed the EMG decomposition technique, which allows the ability to isolate motor unit recruitment during high-load and low-load contractions. Results showed that the higher-load condition recruited a greater portion of high-threshold motor units compared to the lower-load condition. That said, some participants were able to achieve similar recruitment levels regardless of the magnitude of load, indicating recruitment ability may have an individual component. It should be noted that the utility of EMG decomposition for predicting

recruitment has come under scrutiny (62), raising questions about the veracity of these findings.

Tesch and colleagues (253) employed glycogen depletion analysis to evaluate recruitment during dynamic knee extension performance at loads of 30%, 45%, and 60% of 1RM. Results showed that Type IIa fibers began to be recruited at 30% of 1RM, and about half of these fibers showed glycogen loss as the load approached 50% of 1RM. However, the Type IIax and IIx fibers were activated only when the load reached 60% of 1RM. The study was limited by the fact that sets were not carried out to muscular failure. More recently, Morton and colleagues (158) reported similar glycogen depletion in both Type I and Type II fibers during performance of fatiguing heavy (80% of 1RM) versus light (30% of 1RM) loading, indicating that the magnitude of load does not determine recruitment threshold when training is carried out to failure.

Overall, the findings of these studies are somewhat conflicting, but they suggest that the use of light loads is sufficient for recruiting at least a majority of fibers in the available motor unit pool provided training is carried out with a high level of effort. However, recall from chapter 1 that recruitment is but one component for maximizing muscle development; once recruited, the fiber must be stimulated to a sufficient magnitude and for a sufficient period of time. To put things into context, training at a higher percentage of 1RM creates more recruitment and stimulation of higher-threshold motor units upon initiation of a set and trains muscle at an earlier point than when using light loads (99). Alternatively, training with light loads maintains tension in the lower-threshold motor units for an extended period, and buildup of H^+ may interfere with calcium binding in Type II fibers, thereby placing an even greater burden on Type I fibers to maintain force output (86). This could be particularly important in optimizing the development of the Type I fibers, which are highly fatigue resistant. Indeed, an emerging body of research suggests that Type I fiber hypertrophy may be more effective when training with low loads than when training with high loads (85). Moreover, preferential hypertrophy of Type I fibers has been shown in research on low-load blood flow restriction (BFR) training (22, 23,

111), providing further support of a potential fiber type-specific response to load. Indeed, when matched for volitional effort (i.e., sets taken to muscle failure), low-load training and low-load BFR training produce comparable increases in muscle size (7, 67), suggesting the two forms of training may promote hypertrophy through similar mechanisms.

The authors of several animal studies have investigated the acute molecular responses to training at various intensities of load. Using an *in situ* model, Martineau and Gardiner (138) subjected rat plantaris muscles to peak concentric, eccentric, and isometric actions via electrical stimulation. Results showed tension-dependent phosphorylation of JNK and ERK1/2, and higher mechanical tension resulted in progressively greater phosphorylation. This suggests that peak tension is a better predictor of MAPK phosphorylation than either time under tension or rate of tension development. Follow-up work by the same laboratory revealed a linear relationship between time under tension and signaling of JNK, whereas the rate of tension change showed no effect, highlighting the importance of time under tension in anabolic signaling (139). Taken together, these findings point to the importance of overall training volume for maximizing the acute molecular responses related to skeletal muscle hypertrophy irrespective of loading intensity.

Human data provide further insight into the process. Hulmi and colleagues (109) found that early-phase post-exercise MAPK and p70^{S6K} phosphorylation responses were significantly greater following 5 sets of leg press exercises at 10RM compared to 15 sets at 1RM. Taylor and colleagues (247) demonstrated that the ERK1/2 pathway was similarly activated at the upper and lower limits of the medium-repetition range (85% vs. 65% of 1RM), but a strong trend was seen for greater circulating IGF-1 release at the higher intensity of load. Popov and colleagues (182) displayed diverse responses in anabolic signaling and myogenic gene expression following resistance exercise performed at 74% versus 54% of 1RM. Follow-up work from the same lab again showed divergent load-dependent intracellular signal-

ing responses in trained male powerlifters, with greater phosphorylation of p70^{S6K} and 4E-BP1 observed following moderate load (65% of 1RM) training and greater phosphorylation of ERK1/2 seen after high-load training (85% of 1RM) (134).

With respect to muscle protein synthesis, increases in heavy-load and moderate-load training are similar in the initial hours following resistance training on a volume-equated basis (128). In contrast, muscle protein synthesis appears to be blunted at lower intensities of load (<60% of 1RM) when training is not carried out to failure (128). On the other hand, Burd and colleagues (30) reported that muscle protein synthesis over 24 hours was actually greater when training to failure at 30% of 1RM compared to 90% of 1RM. Considering the body of acute data as a whole, findings suggest a robust acute response to resistance training regardless of intensity of load, provided training is carried out with a high intensity of effort and, in the case of heavy loading, volume is equated. However, the responses are complex and suggest a synergism to training across loading zones.

A number of studies have attempted to compare muscular adaptations between loading zones over time. Those investigating heavy versus medium loading have generally favored the hypertrophy range when volume was not equated between groups. Choi and colleagues (48) randomly assigned 11 young men to either a bulk-up protocol of 9 sets of knee extensions at 40% to 80% of 1RM with 30 seconds of rest between sets or a power-up protocol consisting of 5 sets at 90% of 1RM with 3 minutes of rest between sets. After 8 weeks, the results showed significantly greater increases in quadriceps hypertrophy for the bulk-up group. Masuda and colleagues (143) reported similar findings when employing an identical protocol. Alternatively, studies that equated volume between heavy- and medium-load training have failed to demonstrate superiority for the hypertrophy range (39, 47). Taken collectively, the findings suggest that volume, as opposed to intensity of load, was responsible for any observed differences in muscle growth. All of the aforementioned studies used untrained subjects,

limiting the ability to generalize findings to trained lifters.

My lab (208) investigated heavy versus moderate loading in 20 resistance-trained men who were randomly assigned to one of two groups: a hypertrophy group that performed a bodybuilding-style routine, or a strength group that performed a powerlifting-style routine. The hypertrophy group protocol was a split-body routine in which each muscle was worked once per week with 3 exercises per session, performing 3 sets of 10 repetitions and resting 90 seconds between sets. The strength group protocol was a total-body routine in which each muscle was worked 3 times per week with 1 exercise per session, performing 7 sets of 3 repetitions and resting 3 minutes between sets. Volume load was equated so that subjects in both groups lifted approximately the same amount of weight per week. All sets were performed to the point of momentary concentric muscular failure. After 8 weeks, subjects in both groups significantly increased biceps brachii muscle thickness, with no differences between groups. Subjects in both groups also significantly increased 1RM strength, but the strength group had greater increases in the bench press and showed a trend for greater increases in the squat. From a hypertrophy-training standpoint, these results suggest that hypertrophy is similar along a continuum of 3 to 10 repetitions as long as equal volumes are performed, but that maximizing strength requires lifting very heavy weights.

It should be noted that that per-session training time in the strength group was 70 minutes, whereas in the hypertrophy group it was 17 minutes. So, from a time-efficiency standpoint, the bodybuilding-type training produced similar hypertrophy (as well as nearly similar strength increases) in about 1/4 of the time that the powerlifting-type training did. In fact, time constraints associated with the strength group allowed for only three major body areas to be worked in the study: chest (using upper-body pushing exercises), back (using upper-body pulling exercises), and thighs. The efficiency of the hypertrophy group would have allowed for additional volume in the muscle groups trained or the inclusion of exercises for other muscle groups, or both. Working specific muscles

(and aspects of muscles), such as the middle and posterior deltoids, the hamstrings, and the calves, alone would have benefited overall muscle hypertrophy. Moreover, exit interviews revealed that those in the strength group felt overtaxed by the end of the study. Almost all complained of sore joints and general fatigue, and the two dropouts from this group were because of joint-related injury. These results indicate that although mechanistically heavy and moderately heavy weights appear to promote similar hypertrophic responses when volumes are equated, from an application standpoint, it simply is not practical to constantly lift heavy loads at the high volumes needed for maximizing muscle growth.

With respect to high-repetition training, research shows that muscle hypertrophy is diminished when loads higher than 60% of 1RM are not carried out to a point that approaches muscular failure. This was clearly demonstrated in a study by Holm and colleagues (105), in which subjects performed 8 repetitions of knee extensions on one leg and 36 repetitions of knee extensions on the other leg. In the light-load condition, subjects performed 1 repetition every 5th second for 3 minutes, thereby reducing the effects of fatigue; training in the heavy-load condition was carried out in a traditional fashion. Ten sets were performed each session, and training occurred 3 days a week. After 12 weeks, muscle cross-sectional area was 3-fold greater in the group that performed heavy-load training. These findings correlate with acute data showing an attenuation of muscle protein synthesis when training substantially short of failure at intensities of load below 60% of 1RM (128).

Research investigating the hypertrophic effects of light-load training to muscular failure provides compelling evidence that comparable whole-muscle hypertrophy can be achieved over a wide range of loading zones. A recent meta-analysis (214) sought to provide clarity on the topic by evaluating hypertrophic adaptations in randomized experimental trials that compared resistance training at $\leq 60\%$ of 1RM to that at $>60\%$ of 1RM; inclusion criteria required that both conditions were carried out to volitional failure. The pooled data from

10 studies that assessed changes in muscle size using site-specific methods showed no difference between high- versus low-load conditions; the observed effect size difference of 0.03 indicates hypertrophic adaptations were virtually identical. Moreover, findings were consistent across populations including age, sex, and training status. Collectively, the body of literature indicates that load is not a determining factor in the exercise-induced accretion of muscle mass, at least at the whole-muscle level.

The question then arises as to whether there is a minimum loading threshold for eliciting hypertrophic changes. This question was explored in an elegant study by Lasevicius and colleagues (129). Employing a within-subject design, untrained men performed 3 sets of the unilateral leg press and arm curl at 20% of 1RM to volitional failure on one of their limbs. For the other limb, subjects were randomly

assigned to one of three intensities: 40%, 60%, or 80% of 1RM. The 20% of 1RM condition was always trained first in the sequence, and the contralateral limb then performed as many sets as required to achieve the same volume load as attained in the lower-load condition. Results showed almost identical hypertrophy of both the elbow flexors and quadriceps for the 40%, 60%, and 80% of 1RM conditions. However, muscle growth in the 20% condition was less than half that observed with the heavier loads. Taken together with research showing that similar increases in muscle size are obtained with 30% versus 80% of 1RM (153), the findings indicate that an intensity of 30% of 1RM may be a minimum loading threshold for hypertrophy training, below which muscle growth becomes compromised.

Table 4.3 provides a summary of the research related to intensity of load and muscle hypertrophy.

TABLE 4.3 Summary of Hypertrophy Training Studies Investigating Light ($\leq 50\%$ of 1RM) Versus Heavy ($>50\%$ of 1RM) Training Load

Study	Subjects	Design	Study duration	Volume load equated?	Hypertrophy measurement	Findings
Campos et al. (39)	32 untrained young men	Random assignment to high-load (3RM to 5RM), intermediate-load (9RM to 11RM), or low-load (20RM to 28RM) exercise. Exercise consisted of 2 to 4 sets of the squat, leg press, and knee extension, performed 3 days per week. Tempo was consistent between conditions.	8 weeks	Yes	Muscle biopsy (quadriceps)	Significant increases in CSA for high-intensity exercise; no significant increase in CSA for low-intensity exercise. Significantly greater increases in muscle strength for high- vs. low-intensity group.
Franco et al. (73)	32 untrained young women	Random assignment to high-load (8RM to 10RM) or low-load (30RM to 35RM) exercise. Exercise consisted of 3 sets of the leg press, leg extension, and leg curl performed twice per week.	8 weeks	No	DXA	Greater increase in lean soft-tissue mass of the legs favoring the low-load condition
Holm et al. (105)	11 untrained young men	Random, counterbalanced performance of 10 sets of unilateral knee extensions, training one leg at 70% of 1RM and the contralateral leg at 15.5% of 1RM, performed 3 days per week.	12 weeks	Yes	MRI (quadriceps)	Significantly greater increases in quadriceps CSA for high- vs. low-load exercise

Study	Subjects	Design	Study duration	Volume load equated?	Hypertrophy measurement	Findings
Jessee et al. (113)	40 untrained men and women	Within-subject design whereby the legs were randomly assigned to high-load (70% of 1RM) or low-load (15% of 1RM) knee extension exercise. All subjects performed 4 sets carried out twice per week. Rest intervals were 30 and 90 seconds for the low- and high-load conditions, respectively. Note: two additional conditions were investigated using BFR.	8 weeks	No	Ultrasound (quadriceps)	No differences in muscle thickness between conditions
Lasevicius et al. (129)	30 untrained young men	Within-subject design in which subjects performed the unilateral arm curl and leg press in one limb with 20% of 1RM to failure; the other limb was randomly assigned to perform the exercise with either 40%, 60%, or 80% of 1RM. The heavier-load condition was performed with additional sets to equate volume load with the 20% of 1RM condition.	12 weeks	Yes	Ultrasound (quadriceps, elbow flexors)	No differences in CSA changes between 40%, 60%, or 80% of 1RM. CSA increases in the 20% of 1RM condition were significantly lower than the higher-load conditions.
Leger et al. (131)	24 untrained middle-aged men	Random assignment to either high-load (3RM to 5RM) or low-load (20RM to 28RM) exercise. Exercise consisted of 2 to 4 sets of squats, leg presses, and knee extensions, performed 3 days per week.	8 weeks	Yes	CT	No differences in CSA between low- and high-intensity exercise
Lim et al. (132)	21 untrained young men	Random assignment to either high-load (80% of 1RM) to failure, low-load (30% of 1RM) exercise to failure, or low-load (30% of 1RM) exercise not performed to failure but with work matched to the 80% of 1RM condition. Exercise consisted of 3 sets of the leg press, leg extension, and leg curl performed twice per week.	10 weeks	Yes/No	Muscle biopsy (quadriceps)	Greater increases in Type I fiber hypertrophy for 80% and 30% of 1RM to failure than in 30% of 1RM not performed to failure. No differences in Type II fiber hypertrophy between conditions.
Mitchell et al. (153)	18 untrained young men	Random assignment to perform 2 of 3 unilateral knee extension protocols: 3 sets at 30% of 1RM, 3 sets at 80% of 1RM, and 1 set at 80% of 1RM. Tempo was consistent between conditions. Training was carried out 3 days per week.	10 weeks	No	MRI, muscle biopsy (quadriceps)	No differences in CSA between low- and high-intensity exercise. Significantly greater strength gains in high- vs. low-load group.

(continued)

Table 4.3 (continued)

Study	Subjects	Design	Study duration	Volume load equated?	Hypertrophy measurement	Findings
Ogasawara et al. (172)	9 untrained young men	Nonrandomized crossover design to perform 4 sets of bench press exercises at 75% of 1RM. Training was carried out 3 days per week. Tempo was consistent between conditions. After a 12-month washout period, the same protocol was performed at 30% of 1RM.	6 weeks	No	MRI (triceps brachii, pectoralis major)	No differences in CSA between low- and high-intensity exercise. Significantly greater increases in strength favoring high over low load.
Popov et al. (181)	18 untrained young men	Random assignment to either high-intensity (80% of MVC) or low-intensity (50% of MVC) exercise without relaxation. Exercise consisted of leg press exercises performed 3 days per week. Tempo was consistent between conditions.	8 weeks	No	MRI (quadriceps, gluteus maximus)	No differences in CSA between groups
Schoenfeld et al. (210)	18 well-trained young men	Random assignment to a resistance training protocol of either 8RM to 12RM or 25RM to 35RM. All subjects performed 3 sets of 7 exercises. Training was carried out 3 days per week.	8 weeks	No	Ultrasound (elbow flexors, elbow extensors, quadriceps)	No significant differences in thickness of the biceps, triceps, or quadriceps between conditions
Schuenke et al. (221)	34 untrained young women	Random assignment to moderate intensity (80% to 85% of 1RM) at a tempo of 1 to 2 seconds, low intensity (~40% to 60% of 1RM) at a tempo of 1 to 2 seconds, or slow speed (~40% to 60% of 1RM) at a tempo of 10 seconds concentric and 4 seconds eccentric. Exercise consisted of 3 sets of squats, leg presses, and knee extensions, performed 2 or 3 days per week.	6 weeks	No	Muscle biopsy (quadriceps)	Significant increases in CSA for high-intensity exercise; no significant increase in CSA for low-intensity exercise
Stefanaki et al. (240)	13 untrained young women	Within-subject design in which subjects were randomly assigned to perform the arm curl and knee extension at 30% of 1RM in one arm and leg and at 80% of 1RM in the contralateral limbs. Training was carried out twice per week.	6 weeks	No	Ultrasound (elbow flexors, quadriceps)	No differences in muscle thickness between conditions

Study	Subjects	Design	Study duration	Volume load equated?	Hypertrophy measurement	Findings
Tanimoto and Ishii (244)	24 untrained young men	Random assignment to 50% of 1RM with a 6-second tempo and no relaxation phase between reps, 80% of 1RM with a 2-second tempo and 1 second of relaxation between reps, or 50% of 1RM with a 2-second tempo and 1 second of relaxation between reps. Exercise consisted of 3 sets of knee extensions performed 3 days per week.	12 weeks	No	MRI (quadriceps)	No differences in CSA or strength between low- and high-intensity exercise
Tanimoto et al. (245)	36 untrained young men (12 served as non-exercising controls)	Random assignment to either ~55% of 1RM with a 6-second tempo and no relaxation phase between reps or 80% to 90% of 1RM with a 2-second tempo and 1 second of relaxation between reps. Exercise consisted of 3 sets of squats, chest presses, lat pulldowns, abdominal bends, and back extensions, performed 2 days per week.	13 weeks	No	Ultrasound (chest, anterior and posterior upper arm, abdomen, subscapular, anterior and posterior upper thigh)	No differences in CSA or strength between low- and high-intensity exercise
Van Roie et al. (258)	56 untrained older men and women	Random assignment of leg press and knee extension training at high load (2 × 10 to 15 reps at 80% of 1RM), low load (1 × 80 to 100 reps at 20% of 1RM), or low-load+ (1 × 60 reps at 20% of 1RM, followed by 1 × 10 to 20 reps at 40% of 1RM). Tempo was consistent between conditions.	12 weeks	No	CT (quadriceps)	No differences in muscle volume between groups. Greater increases in strength for high- and low- vs. low-load conditions.
Weiss et al. (266)	44 untrained young men	Random assignment to high-load (3RM to 5RM), moderate-load (13RM to 15RM), or light-load (23RM to 25RM) resistance training. Exercise consisted of 3 sets of squats performed 3 days per week.	7 weeks	No	Ultrasound (quadriceps, hamstrings)	No significant differences in muscle thickness between conditions

Abbreviations: RM = repetition maximum; CSA = cross-sectional area; CT = computerized tomography; MRI = magnetic resonance imaging; BFR = blood flow restriction; DXA = dual X-ray absorptiometry; MVC = maximum voluntary contraction.

PRACTICAL APPLICATIONS**LOAD**

Hypertrophy can be achieved across a wide spectrum of loading zones, with no differences apparent at the whole-muscle level. Low-load training emphasizes metabolic stress and promotes the greatest increases in local muscular endurance, whereas low-repetition, high-load training requires high levels of mechanical tension and enhances the ability to lift heavier loads as a result of greater neural adaptations. Some evidence suggests there may be a fiber type–specific response in which heavy-load training produces greater cross-sectional area increases in Type II fibers, and light loads have a preferential effect on Type I hypertrophy. Thus, if the primary goal is maximizing hypertrophy without regard to strength-related factors, then training across a wide spectrum of repetition ranges (1 through 20+) is recommended to exploit all possible avenues for the complete development of the whole muscle. There is merit to focusing on a medium-repetition range (6RM to 12RM) because it provides high levels of mechanical tension sufficient to stimulate the full array of fiber types while allowing for sufficient training volumes. Incorporating heavy loading (1RM to 5RM) enhances strength, which ultimately allows the use of heavier loads during medium-repetition lifting. Additionally, light-load training should be included both to ensure the optimal development of Type I fibers and to improve the buffering capacity of muscle so that additional repetitions can be performed at a given medium intensity of load. Training below approximately 30% of 1RM may be insufficient to fully stimulate optimal muscular development, although its potential ability to selectively target development of Type I fibers remains to be determined.

On the other hand, if the goal is to promote hypertrophy to maximize muscular strength, there appears little reason to employ loads less than approximately 70% of 1RM, other than perhaps during deload periods. The compelling body of research indicates the presence of a strength–endurance continuum, in which lighter loads promote the ability to carry out submaximal resistive efforts at the expense of maximal force production (39).

Exercise Selection

The human body is designed to carry out movement in three-dimensional space. Muscle architecture is intricately arranged to accomplish complex movement patterns efficiently and effectively. Therefore, varying exercise parameters (i.e., angle of pull, plane of movement, position of extremities) can preferentially target aspects of the musculature, as well as make synergists and stabilizers more active or less active (205). Thus, choice of exercise may contribute to the degree of selective hypertrophy of specific muscles (91).

Numerous muscles have common origins, but their fibers diverge to insert at different attachment sites. These different heads provide greater leverage for carrying out multiplanar movement. A classic example is the deltoid muscle: The anterior deltoid performs shoulder flexion, the middle deltoid performs abduction, and the posterior deltoid performs horizontal abduction. Other examples are the pectoralis major (clavicular and sternal heads), biceps brachii (short and long heads), and gastrocnemius (medial and lateral heads). Moreover, the direction of the fibers in a given muscle allow for greater or lesser leverage in a given move-

ment. The trapezius, for example, is subdivided so that the upper aspect elevates the scapula, the middle aspect abducts the scapula and the lower aspect depresses the scapula (133).

Evidence suggests that it is possible to target not only different aspects of a muscle but also portions of a given muscle fiber as a result of *fiber partitioning*. The partitioning hypothesis is based on research showing that the arrangement of individual muscles is more complex than simply a bundle of fibers attaching at aponeuroses, tendons, or bones with a single muscle nerve innervation (61). Rather, many muscles are segmented into distinct compartments, and these compartments are innervated by their own neural branches. Muscles such as the sartorius, gracilis, semitendinosus, and biceps femoris contain subdivisions of individual fibers that are innervated by separate motor neurons (271, 273). Moreover, the sartorius and gracilis, among other muscles, are actually composed of relatively short, in-series fibers that terminate intrafascicularly, refuting the supposition that myofibers always span the entire origin to insertion (101).

Muscular partitions may have functional or task-oriented roles; that is, different portions of one muscle may be called into play depending on the task-relevant demands of the situation (61). This is exemplified in the biceps brachii, in which both the long and short heads have architectural compartments that are innervated by private branches of the primary neurons (223). Research indicates that fibers in the lateral portion of the long head of the muscle are recruited for elbow flexion, fibers in the medial aspect are recruited for supination, and fibers that are centrally located are recruited for non-linear combinations of flexion and supination (250, 251). Moreover, the short head demonstrates greater activity in the latter part of an arm curl (i.e., greater elbow flexion), whereas the long head is more active in the early phase of movement (28). These findings lend support to the notion that a variety of exercises will ensure the complete stimulation of all fibers.

Although evidence that varying the exercises enhances muscle activation is compelling, the extent to which selective activation of a given portion of a muscle enhances its site-specific

hypertrophic response remains to be determined. A large body of research shows that muscle hypertrophy occurs in a nonuniform fashion in terms of preferential growth of both individual muscles in a muscle group and different regions within the same muscle. For example, multiple studies have shown that knee extension exercises result in a heterogeneous hypertrophic response in which certain areas of the quadriceps femoris show greater hypertrophy than others (91, 108, 163). Similar nonuniform growth has been demonstrated in the triceps brachii following regimented elbow extension exercises (262, 263).

Some evidence suggests that regional hypertrophy is specific to the site of muscle activation. Using magnetic resonance imaging technology, Wakahara and colleagues (262) determined muscle activation in a group of subjects performing 5 sets of 8 repetitions of the lying triceps extension exercise. Another group of subjects then underwent a 12-week supervised exercise program employing the same variables used in the acute activation study. Results showed that the extent of hypertrophy in the triceps was specific to the region of activation. Follow-up work by the same lab showed a similar outcome from the close-grip bench press exercise; triceps hypertrophy correlated to the site of activation, but occurred in a different region of the muscle compared to the previous study (263). To the contrary, other research shows that regional differences in quadriceps femoris hypertrophy following regimented resistance training are a function of muscle oxygenation status during exercise as opposed to neuromuscular activity (154).

Fonseca and colleagues (71) demonstrated the importance of varying exercise selection in a study in which they compared muscular adaptations following performance of the Smith machine squat with a volume-equated combination of the Smith machine squat, leg press, lunge, and deadlift. Results showed that the varied exercise routine produced more uniform muscle hypertrophy of all four quadriceps muscles compared to performing the Smith machine squat alone. In fact, the Smith machine squat failed to significantly increase cross-sectional area in the vastus medialis and

KEY POINT

Once trainees have learned the movement patterns of basic resistance training exercises, they should use a variety of exercises to maximize whole-body muscle hypertrophy. This should include free-form as well as machine-based exercises. Similarly, both multi- and single-joint exercises should be included in hypertrophy-specific routines to maximize muscular growth.

rectus femoris muscles. It is interesting to speculate whether hypertrophic results would have been enhanced even further if more targeted single-joint exercises, such as the knee extension, had been included in the varied routine because research indicates this movement preferentially targets the rectus femoris (59, 63).

Although the growth-related benefits of training variety are clear, the concept should not be taken to an extreme. When exercise variation occurs too frequently, a person may spend too much time developing motor skills with suboptimal loads, which compromises

the hypertrophic response (99). This is particularly important during the initial stages of training in which improvements in strength are largely related to an improved neuromuscular response (see chapter 1). During this motor learning period, the number of exercises in a program should be limited so that neural patterns become ingrained into the subconscious. On the other hand, trained lifters can be more liberal in varying exercise selection; their neural patterns are much more entrenched, and depending on the complexity of the exercise, coordinated movements are maintained even after a lengthy period without training. Moreover, significant transfer of training from exercise variations (e.g., back squat to front squat) facilitates the retention of neural patterns over time. As a general rule, more complex movements, particularly those involving multi-joint free weight exercise, should be maintained in a regular program rotation while those requiring less skill to perform (e.g., single-joint and machine exercises) can be varied more freely.

Table 4.4 provides a summary of the research related to exercise selection and muscle hypertrophy.

TABLE 4.4 Summary of Hypertrophy Training Studies Investigating Exercise Selection

Study	Subjects	Design	Study duration	Volume equated?	Hypertrophy measurement	Findings
Balsalobre et al. (11)	21 resistance-trained young men	Random assignment to a resistance training protocol using either a fixed exercise selection or having exercises randomly varied each session via a computerized app. Both groups performed 3 sets of 6 exercises, with training carried out 4 times per week.	8 weeks	Yes	Ultrasound (quadriceps)	No significant differences in muscle thickness between conditions but the magnitude of changes modestly favored the fixed exercise selection condition
Fonseca et al. (71)	49 untrained young men	Random assignment to a resistance training protocol involving performance of the Smith machine squat or a combination of the Smith machine squat, leg press, lunge, and deadlift. All subjects performed the routine twice per week at 6RM to 10RM for each exercise.	12 weeks	Yes	MRI (quadriceps)	Greater hypertrophy of the vastus medialis and rectus femoris muscles in the varied-exercise condition

Abbreviation: RM = repetition maximum; MRI = magnetic resonance imaging.

PRACTICAL APPLICATIONS**EXERCISE SELECTION**

Architectural variances of individual muscles lend support to the notion of the need to adopt a multiplanar, multiangled approach to hypertrophy training using a variety of exercises. Moreover, evidence suggests that regular exercise rotation is warranted to fully stimulate all fibers within a muscle and thus maximize the hypertrophic response.

As mentioned in chapter 1, neural mechanisms are primarily responsible for increases in strength during the early stages of resistance training. Thus, lifters in the initial training phase should focus on acquiring the necessary motor learning and control to effectively carry out exercise performance. Simplification and repeated practice are important in this context. Performing the same movements over and over ingrains motor patterns so that proper technique becomes second nature. For those who have difficulty with coordination, reducing degrees of freedom with machine-based training can be an effective means to enhance neural development. They can then progress to more complex variations in three-dimensional space.

A variety of exercises should be employed over the course of a periodized training program to maximize whole-body muscle hypertrophy, with a particular focus on working muscles based on their anatomical design. This should include a combination of free-form exercises (i.e., free weights and cables) that maximize the contribution of stabilizer muscles, as well as machine-based movements that target specific muscles or portions thereof. Similarly, both multi- and single-joint exercises should be included in a hypertrophy-specific routine to maximize muscular growth.

Type of Muscle Action

Mechanosensors are sensitive not only to the magnitude and duration of stimulation, but also to the type of imposed action. As discussed in chapter 1, the three basic types of muscle actions are concentric, eccentric, and isometric. Mechanistically, there is a logical basis for speculation that eccentric actions produce the greatest anabolic response, and research often focuses on this type of muscle action. Eccentric strength is approximately 20% to 50% greater than concentric strength (12) and allows heavier loading during exercise. Moreover, forces generated during eccentric training are approximately 45% higher than those generated during concentric training (115) and approximately double that of isometric contractions (205). The greater mechanical tension per active fiber is thought to be due to a reversal of the size principle of recruitment, whereby Type II fibers are selectively recruited

at the expense of Type I fibers (228, 242). Evidence for preferential Type II recruitment has been noted during plantar flexion, as has derecruitment of the slow-twitch soleus muscle and the corresponding increase in activity of the gastrocnemius during the eccentric component of movement (162). These findings are consistent with EMG data indicating selective recruitment of a small number of motor units during eccentric hamstring exercise, including additional recruitment of previously inactive motor units (147). However, other research shows that Type I and Type II fibers are equally glycogen depleted following eccentric exercise, suggesting no preferential recruitment of high-threshold motor units (248).

Hypertrophic advantages of eccentric exercise are also thought to be related to muscle damage (206). Although concentric and isometric exercise can induce muscle damage, the extent of damage is heightened during eccentric actions. This is believed to be due to

greater force demands on fewer active fibers, which are prone to tear when attempting to resist lengthening. Because the weakest sarcomeres are located at different regions of each myofibril, it is hypothesized that the associated nonuniform lengthening causes a shearing of myofibrils. This deforms membranes, particularly T-tubules, leading to a disturbance of calcium homeostasis that further damages muscle tissue by eliciting the release of the calcium-activated neutral proteases involved in degrading Z-line proteins (4, 18). However, muscle damage is substantially attenuated over time via the repeated bout effect (148); thus, the implications in this regard are questionable for well-trained lifters, although it is possible that relatively modest myofiber disruptions from eccentric training may confer a hypertrophic effect. It also is conceivable that early-phase myodamage may induce greater satellite cell responses that ultimately drive greater long-term growth. The relevance of muscle damage in eccentric training-related hypertrophic adaptations remains speculative.

A number of researchers have investigated the acute signaling response to modes of contractions. Franchi and colleagues (72) found that eccentric training preferentially upregulated early MAPK activation (p38 MAPK, ERK1/2, p90^{RSK}) compared to concentric training, but neither mode affected Akt/mTOR or inflammatory signaling 30 minutes after exercise. Eliasson and colleagues (60) found that maximal eccentric actions (4 sets of 6 repetitions) significantly increased early-phase (2 hours) phosphorylation of p70^{S6K} and the ribosomal protein S6, whereas the same number of maximal concentric actions showed no effect on phosphorylation of these signaling molecules. Consistent with the study by Franchi and colleagues, neither contraction mode produced significant increases in Akt or mTOR, suggesting that eccentric actions activate p70^{S6K} via an Akt-independent pathway. In addition, eccentric exercise was shown to promote significantly greater upregulation of STARS mRNA compared to concentric exercise (10-fold vs. 3-fold, respectively) as well as greater expression of downstream serum

response factor (SRF) target genes (261). These findings suggest that eccentric exercise preferentially modulates the transcription of specific myofibrillar genes associated with hypertrophic adaptations to resistance exercise, possibly as a mechanism to protect against contractile-induced muscle damage.

Research investigating the effect of contraction modes on muscle protein synthesis has produced disparate results. Several studies have failed to demonstrate any differences in either mixed muscle (78, 177) or myofibrillar (52) muscle protein synthesis after submaximal eccentric or concentric resistance exercise. Conversely, Moore and colleagues (157) reported a more rapid rise in myofibrillar muscle protein synthesis following 6 sets of 10 work-matched maximal eccentric versus concentric knee extension repetitions. The discrepancies between findings suggest that although muscle protein synthesis is similar in all contraction modes during submaximal exercise, maximal eccentric actions enhance the accretion of muscle proteins.

Longitudinal studies provide limited evidence of a hypertrophic advantage from eccentric actions. In a meta-analysis encompassing 15 studies meeting inclusion criteria, our lab (216) found a superior effect size difference for hypertrophic outcomes favoring eccentric compared to concentric actions (ES = 0.27), with results translating into modestly greater increases in muscle growth across studies (10.0% vs. 6.8%, respectively). These results must be taken in the context that the majority of studies included for analysis matched total repetitions as opposed to total work. Considering that maximal eccentric strength is greater

KEY POINT

Concentric and eccentric muscle actions appear to recruit muscle fibers in different orders, result in different signaling responses, and produce distinct morphological adaptations in muscle fibers and fascicles. Therefore, both concentric and eccentric actions should be incorporated during training to maximize the hypertrophic response.

than concentric strength, it is possible that observed differences in hypertrophy between conditions may in fact be attributed to a greater amount of work performed during eccentric actions. Limited research into matching total work between conditions provides somewhat discrepant findings. In one study (96), measures of midthigh lean mass were only increased in those performing eccentric actions, whereas concentric training showed no significant changes. Alternatively, Moore and colleagues (156) found a slight hypertrophic benefit to eccentric versus concentric training under work-matched conditions (6.5% vs. 4.6%), but results did not reach statistical significance. Emerging evidence suggests that adding supra-maximal eccentric actions to a training program may enhance hypertrophic adaptations (202); see chapter 5 for further discussion on the topic.

One thing that appears quite clear from the literature is that concentric and eccentric actions produce distinct morphological adaptations at the fiber and fascicle levels. Franchi and colleagues (72) found that eccentric training produced significantly greater increases in

fascicle length compared to concentric training (12% vs. 5%, respectively), whereas concentric actions produced significantly greater increases in pennation angle (30% vs. 5%). These findings are consistent with those of other research on the topic (190, 227) and indicate a predisposition toward in-series hypertrophy following eccentric exercise. Interestingly, fascicle length changes seem to be specific to the initial stages of resistance training; increases abate after 5 weeks of consistent training (24).

Contraction modes also display region-specific effects on hypertrophy; eccentric actions show preferential growth in the distal aspect of the vastus lateralis (8% eccentric vs. 2% concentric), and concentric actions target the midportion of the muscle (7% eccentric vs. 11% concentric) (72). It is speculated that site-specific hypertrophy might be related to regional muscle damage along the length of the fiber that consequently results in nonuniform changes in muscle activation (98).

Table 4.5 provides a summary of the research related to type of muscle action and muscle hypertrophy.

TABLE 4.5 Summary of Hypertrophy Training Studies Investigating Type of Muscle Action

Study	Subjects	Design	Study duration	Work matched?	Mode	Hypertrophy measurement	Findings
Ben-Sira et al. (19)	48 untrained young women	Random assignment to a resistance training protocol of eccentric-only, concentric-only, mixed eccentric and concentric, or supra-maximal eccentric actions for the knee extensors. The mixed-condition group performed 3 sets of 10 reps at 65% of concentric 1RM; the supra-maximal eccentric group performed 3 sets of 5 reps at 130% of concentric 1RM; the concentric-only and eccentric-only groups performed 3 sets of 10 reps for these actions at 65% of concentric 1RM. Training was carried out twice per week.	8 weeks	No	Knee extension machine	Circumference measurement	No significant differences in thigh girth between conditions

(continued)

Table 4.5 (continued)

Study	Subjects	Design	Study duration	Work matched?	Mode	Hypertrophy measurement	Findings
Blazevich et al. (24)	21 untrained young men and women	Random assignment to a resistance training protocol of either eccentric or concentric actions for the knee extensors. All subjects performed 4 to 6 sets of 6 maximal reps. Training was carried out 3 days per week.	10 weeks	Yes	Isokinetic dynamometer	MRI, ultrasound (quadriceps)	No significant differences in quadriceps hypertrophy between conditions
Cadore et al. (35)	22 recreationally trained young men and women	Random assignment to a resistance training protocol of either eccentric or concentric actions for the knee extensors. All subjects performed 2 to 5 sets of 8 to 10 maximal reps. Training was carried out twice weekly.	6 weeks	Yes	Isokinetic dynamometer	Ultrasound (quadriceps)	No significant differences in muscle thickness between conditions
Farup et al. (66)	22 untrained young men	Within-subject design in which subjects performed concentric actions of the knee extensors with one leg and eccentric actions with the other leg. All subjects performed 6 to 12 sets of 6RM to 15RM. Eccentric actions were performed at 120% of concentric 1RM. Training was carried out 3 days per week.	12 weeks	No	Knee extension machine	MRI (quadriceps)	No significant differences in quadriceps hypertrophy between conditions
Farup et al. (65)	22 untrained young men	Within-subject design in which subjects performed concentric actions of the knee extensors with one leg and eccentric actions with the other leg. All subjects performed 6 to 12 sets of 6RM to 15RM. Eccentric actions were performed at 120% of concentric 1RM. Training was carried out 3 days per week.	12 weeks	No	Knee extension machine	Muscle biopsy (quadriceps)	Significantly greater increases in Type II fiber CSA for the concentric condition

Study	Subjects	Design	Study duration	Work matched?	Mode	Hypertrophy measurement	Findings
Farthing and Chilibeck (64)	36 untrained young men and women	Within-subject design in which subjects performed concentric actions of the elbow flexors with one arm and eccentric actions with the other arm. Subjects were randomly assigned to perform the actions at either a fast or slow speed. All subjects performed 2 to 6 sets of 8 maximal reps. Training was carried out 3 days per week.	8 weeks	Yes	Isokinetic dynamometer	Ultrasound (elbow flexors)	Greater increase in muscle thickness for the eccentric condition
Franchi et al. (72)	12 untrained young men	Random assignment to a resistance training protocol of either eccentric or concentric actions of the lower-limb extensors. All subjects performed 4 sets of 8RM to 10RM. Eccentric actions were performed at 120% of concentric 1RM. Concentric actions were performed for 2 seconds and eccentric actions for 3 seconds. Training was carried out 3 days per week.	10 weeks	No	Leg press machine	MRI (quadriceps)	No significant differences in thigh hypertrophy between conditions
Higbie et al. (102)	54 untrained young women	Random assignment to a resistance training protocol of either eccentric or concentric actions for the knee extensors. All subjects performed 3 sets of 10 maximal reps. Training was carried out 3 days per week.	10 weeks	Yes	Isokinetic dynamometer	MRI (quadriceps)	Significantly greater increases in quadriceps muscle hypertrophy for the eccentric condition
Hortobagyi et al. (106)	21 untrained young men	Random assignment to a resistance training protocol of either eccentric or concentric actions for the knee extensors. All subjects performed 4 to 6 sets of 8 to 12 maximal reps. Training was carried out 3 days per week.	12 weeks	Yes	Isokinetic dynamometer	Muscle biopsy (quadriceps)	Significantly greater increase in Type II fiber hypertrophy of the quadriceps for the eccentric condition
Hortobagyi et al. (107)	48 untrained young men and women	Random assignment to a resistance training protocol of either eccentric or concentric actions for the knee extensors. All subjects performed 4 to 6 sets of 8 to 12 maximal reps. Training was carried out 3 days per week.	12 weeks	Yes	Isokinetic dynamometer	Muscle biopsy (quadriceps)	Significantly greater increase in quadriceps hypertrophy of all fiber types for the eccentric condition

(continued)

Table 4.5 (continued)

Study	Subjects	Design	Study duration	Work matched?	Mode	Hypertrophy measurement	Findings
Jones and Rutherford (115)	12 untrained young men and women	Within-subject design in which subjects performed concentric actions of the knee extensors with one leg and eccentric actions with the other leg. All subjects performed 4 sets of 6 maximal reps. Eccentric actions were performed at 145% of concentric 1RM. Training was carried out 3 days per week.	12 weeks	No	Variable resistance knee extension machine	CT (quadriceps)	No significant differences in thigh hypertrophy between conditions
Kim et al. (121)	13 young men and women (training status not disclosed)	Random assignment to a resistance training protocol of either eccentric or concentric actions for the shoulder abductors. All subjects performed 4 to 6 sets of 6 to 8 maximal reps. Training was carried out 3 days per week.	8 weeks	Yes	Isokinetic dynamometer	Ultrasound (supraspinatus)	No significant differences in hypertrophy of the supraspinatus between conditions
Komi and Buskirk (122)	31 untrained young men	Random assignment to a resistance training protocol of either eccentric or concentric actions of the forearm flexors. Training was carried out 4 days per week.	7 weeks	No	Isotonic dynamometer	Circumference measurements	Greater increases in upper-arm girth for the eccentric condition
Maeo et al. (137)	12 untrained young men	Within-subject design whereby subjects performed concentric actions of the knee extensors with one leg and eccentric actions with the other leg. All subjects performed 3 to 6 sets of maximal reps. Training was carried out twice per week.	10 weeks	Yes	Isokinetic dynamometer	MRI (quadriceps)	Muscle volume increased only in the eccentric condition
Mayhew et al. (144)	20 untrained young men and women	Random assignment to a resistance training protocol of either eccentric or concentric actions for the knee extensors. Concentric actions were performed at an intensity of 90% of maximal concentric power, whereas eccentric actions were performed at the same relative power level. Training was carried out 3 days per week.	4 weeks	No	Isokinetic dynamometer	Muscle biopsy (quadriceps)	Greater increases in Type II quadriceps fiber area for the concentric condition

Study	Subjects	Design	Study duration	Work matched?	Mode	Hypertrophy measurement	Findings
Moore et al. (156)	9 untrained young men	Within-subject design in which subjects performed concentric actions of the elbow flexors with one arm and eccentric actions with the other arm. All subjects performed 2 to 6 sets of 10 maximal reps. Training was carried out twice per week.	9 weeks	Yes	Isokinetic dynamometer	CT (elbow flexors)	No significant differences in hypertrophy of the elbow flexors between conditions
Nickols-Richardson et al. (167)	70 untrained young women	Random assignment to a resistance training protocol of either eccentric or concentric actions for the limbs. All subjects performed 5 sets of 6 maximal reps. Training was carried out 3 days per week.	5 months	No	Isokinetic dynamometer	DXA	No significant differences in fat-free soft-tissue mass between conditions
Reeves et al. (190)	19 untrained elderly men and women	Random assignment to a resistance training protocol of either eccentric or mixed (eccentric and concentric) actions for the lower body. The mixed condition was performed for 2 sets of 10 reps with a load of ~80% of the mixed-action 5RM. The eccentric-only condition was performed for 2 sets of 10 reps with a load of ~80% of the eccentric 5RM. Training was carried out 3 days per week.	14 weeks	No	Knee extension and leg press machines	Ultrasound (quadriceps)	No significant differences in vastus lateralis thickness between conditions
Ruas et al. (199)	40 untrained young men	Random assignment to 1 of 4 protocols: concentric quadriceps/concentric hamstrings; eccentric quadriceps/eccentric hamstrings; concentric quadriceps/eccentric hamstrings; or no-training control. Volume began at 1 set and was progressively increased by adding an additional set each week. Training was performed twice per week.	6 weeks	No	Isokinetic dynamometer	Ultrasound (quadriceps, hamstrings)	Greater increases in muscle thickness for concentric/eccentric and eccentric/eccentric conditions vs. concentric/concentric

(continued)

Table 4.5 (continued)

Study	Subjects	Design	Study duration	Work matched?	Mode	Hypertrophy measurement	Findings
Seger et al. (224)	10 untrained young men	Within-subject design in which subjects performed concentric actions of the knee extensors with one leg and eccentric actions with the other leg. All subjects performed 4 sets of 10 maximal reps. Training was carried out 3 days per week.	10 weeks	No	Isokinetic dynamometer	MRI (quadriceps)	Greater increases in whole-quadriceps muscle hypertrophy distally for the eccentric condition
Smith and Rutherford (233)	10 untrained young men and women	Within-subject design in which subjects performed concentric actions of the knee extensors with one leg and eccentric actions with the other leg. All subjects performed 4 sets of 10 maximal reps. Eccentric actions were performed at 135% of concentric 1RM. Training was carried out 3 days per week.	20 weeks	No	Knee extension machine	CT (quadriceps)	No significant differences in quadriceps hypertrophy between groups
Vikne et al. (259)	17 resistance-trained young men	Random assignment to a resistance training protocol of either eccentric or concentric actions for the elbow flexors. Training was divided between maximum and medium days. Those in the maximum training group performed 3 to 5 sets of 4RM to 8RM; those in the medium training group performed 3 or 4 sets of the same repetition scheme but with lighter loads. Concentric actions were performed explosively, whereas eccentric actions were performed in 3 to 4 seconds. Training was carried out 2 or 3 days per week.	12 weeks	No	Specially designed cable pulley apparatus	CT, biopsy (elbow flexors)	Significantly greater increases in whole-muscle CSA of the elbow flexors for the eccentric condition. Greater increases in Type I and Type II fiber area for the eccentric condition.

Abbreviations: RM = repetition maximum; CSA = cross-sectional area; CT = computerized tomography; MRI = magnetic resonance imaging; DXA = dual X-ray absorptiometry.

PRACTICAL APPLICATIONS

TYPE OF MUSCLE ACTION

Both concentric and eccentric actions should be included in hypertrophy-oriented training programs. These actions appear to complement each other from a growth standpoint. There is a lack of research investigating whether isometric actions provide an additive hypertrophic benefit when combined with dynamic concentric and eccentric training.

Rest Interval Length

The time taken between sets is referred to as the *rest interval*, or *rest period*. Rest intervals can be classified into three broad categories: short (30 seconds or less), moderate (60 to 90 seconds), and long (3 minutes or more) (205). Research demonstrates that rest interval length has distinct effects on the acute response to resistance training, and these responses have been hypothesized to affect chronic hypertrophic adaptations.

Short rest intervals have been shown to markedly increase metabolite accumulation. Ratamess and colleagues (189) found that 30-second rest intervals reduced training volume by more than 50% over the course of 5 sets at 10RM, and marked decreases in load were seen in each subsequent set. Thus, metabolic enhancement is achieved at the expense of reduced mechanical tension, resulting in the need to progressively reduce the amount of loading over subsequent sets to sustain performance in a given repetition range.

Long rest intervals provide a sustained ability to maintain mechanical tension throughout each successive set. Strength capacity has been shown to be largely preserved over 3 sets with rest intervals of 3 minutes or more (123, 189). However, metabolite accumulation diminishes with increasing rest between sets, particularly with respect to lactic acid buildup (1).

Moderate rest periods are believed to provide an ideal compromise between metabolic stress and mechanical tension. A hypertrophy-type workout in which people rested 90 seconds between sets showed significantly greater increases in blood lactate concentration and reductions in pH compared to a strength-type workout with 5 minutes of rest between sets (166). With respect to the

effect on loading, Medeiros and colleagues (152) found that using 60-second rest intervals required a reduction of 5% to 10% in each successive set to allow for the maintenance of 8RM to 12RM loads in resistance-trained subjects. Because moderate rest intervals induce a favorable metabolic environment without substantially compromising mechanical forces, a rest interval of 60 to 90 seconds is often prescribed for maximizing hypertrophy.

Despite the commonly accepted belief that hypertrophy-oriented routines benefit from moderate rest between sets, only a handful of studies have directly investigated the effect of rest intervals on muscle growth over time. The results of these studies have been conflicting, with some showing a potential benefit from longer rest intervals (34, 211), others showing a potential benefit from shorter rest (260), and yet others showing no differences between conditions (3, 69). Differences in findings may be related to the methods and populations studied. For example, studies have included untrained young men (34, 69, 178), trained young men (3, 211), untrained older men (260), and untrained women (103). It remains unclear how the length of rest intervals may affect these diverse populations.

Moreover, only three studies to date have assessed hypertrophy using a site-specific method. Employing a crossover design in resistance-trained men, Ahtiainen and colleagues (3) reported no differences in MRI-derived quadriceps cross-sectional area between 2-minute and 5-minute rest periods. Conversely, a study from my lab found superior increases in muscle thickness via B-mode ultrasound favoring 3-minute over 1-minute rest intervals in a cohort of resistance-trained men (211). Further confounding matters, Fink and colleagues (69) showed that

changes in triceps brachii cross-sectional area as measured by MRI were similar between rest intervals of 30 and 150 seconds during training carried out using light loads (40% of 1RM) to failure; however, modestly greater (although statistically nonsignificant) increases were observed for the thigh muscles when training with longer compared to shorter rest intervals (8.3% vs. 5.7%, respectively).

As with any resistance-training variable, rest intervals can be varied over a given training cycle. In a novel research design, de Souza and colleagues (56) randomized 20 resistance-trained men to either a group that used a constant rest interval or a group that used descending rest intervals. All of the men began by performing 3 sets of 10 to 12 repetitions with 2 minutes of rest for the first 2 weeks. Thereafter, the length of the rest interval progressively decreased to 30 seconds in the descending rest interval group over an ensuing 6-week period, whereas that of the constant rest interval group remained the same throughout the study. After 8 weeks, both groups had significantly increased hypertrophy of the upper and lower extremities; no significant differences were noted in rest interval conditions despite a reduction in training volume for the descending group. A follow-up study using essentially the same protocol but with subjects receiving creatine supplementation again found no significant hypertrophic differences between constant and descending rest intervals (236). Interestingly, effect sizes were substantially greater for descending versus constant rest intervals in the cross-sectional area of both the upper arm (2.53 vs. 1.11, respectively) and thigh (3.23 vs. 2.02, respectively).

Given the highly heterogeneous populations and methodological designs of the studies, evidence on the topic is difficult to fully reconcile. A case can be made that somewhat longer rest intervals (≥ 2 minutes) are preferable for hypertrophy-oriented training because this approach helps to preserve volume load across sets. Unpublished findings from our group found greater increases in quadriceps cross-sectional area as well as higher volume loads for longer (3 minutes) versus shorter (1 minute) rest intervals when the number of sets were equated between conditions. However, performing additional sets in the shorter rest condition to equate volume

KEY POINT

Although rest periods of 60 to 90 seconds induce a seemingly favorable metabolic environment for achieving hypertrophy, research indicates that resting at least 2 minutes between sets provides a hypertrophic advantage compared to shorter rest periods because of the ability to maintain greater volume load.

load resulted in similar hypertrophic changes, thereby highlighting the importance of considering volume load when choosing an appropriate rest interval.

It should be noted that research studies on rest intervals generally have involved protocols in which subjects trained to volitional failure in all sets. This level of exertion necessarily generates greater fatigue than nonfailure sets, thus affecting interset recovery and, in turn, volume load. Higher levels of effort therefore require longer rest intervals to maintain volume load, while stopping short of failure allows for the use of shorter rest periods without compromising volume load.

Exercise selection is another factor worthy of consideration when determining rest intervals for hypertrophy-oriented goals. Multi-joint exercises, particularly those employing free weights, cause substantially more fatigue than single-joint exercises. This was elegantly demonstrated in a study showing a much larger drop-off in the number of repetitions completed over the course of 5 sets during performance of the bench press compared to the machine chest fly in a cohort of resistance-trained men; in fact, the chest fly showed almost no reduction in repetitions over the first 3 sets in the series (225). It can therefore be inferred that while multi-joint free weight exercises require rest periods of at least 2 minutes, somewhat shorter rest periods can be used during performance of single-joint exercises. Such an approach conceivably can take advantage of the greater metabolic disturbances associated with shorter rest during training without compromising volume load.

Table 4.6 provides a summary of the research related to rest interval length and muscle hypertrophy.

TABLE 4.6 Summary of Hypertrophy Training Studies Investigating Rest Interval Length

Study	Subjects	Design	Study duration	Volume load equated?	Hypertrophy measurement	Findings
Ahtiainen et al. (3)	13 resistance-trained young men	Within-subject crossover design in which all subjects performed a resistance training protocol with either 2 or 5 minutes of rest between sets. Training consisted of a multiset split-body routine of 8 to 12 reps carried out 4 days per week.	12 weeks	No	MRI (quadriceps)	No significant differences seen in muscle CSA between groups
Buresh et al. (34)	12 untrained young men	Random assignment to a resistance training protocol with either 1 or 2.5 minutes of rest between sets. Training consisted of a multiset split-body routine of 8 to 11 reps carried out 4 days per week.	10 weeks	No	Hydrostatic weighing, circumference measurements	Significantly greater increases in arm CSA and a trend for greater increases in thigh CSA for the longer rest interval condition
Fink et al. (69)	21 untrained young men	Random assignment to a resistance training program with either 30-second or 150-second rest intervals. Exercise consisted of 4 sets of the squat and bench press at 40% of 1RM performed to failure 2 days per week.	8 weeks	No	MRI (triceps, thigh)	No significant differences noted in CSA measurements between conditions. Relative differences for increases in thigh CSA favored the longer rest condition.
Hill-Haas et al. (103)	18 untrained women	Random assignment to a total-body resistance training program with either 20-second or 80-second rest intervals. Training consisted of 2 to 5 sets of 15RM to 20RM carried out 3 days per week.	5 weeks	No	Circumference measurements (thigh)	No significant differences in thigh circumference between conditions. Relative increases favored the shorter-rest group.
Piirainen et al. (178)	21 untrained young men	Random assignment to a total-body resistance training program with rest intervals based either on individual heart rate (~55 seconds) or a fixed 120-second rest interval. Training consisted of 3 sets of 10RM to 15RM carried out 3 days per week.	7 weeks	No	BIA	No significant differences in FFM between conditions

(continued)

Table 4.6 (continued)

Study	Subjects	Design	Study duration	Volume load equated?	Hypertrophy measurement	Findings
Schoenfeld et al. (204)	21 resistance-trained young men	Random assignment to a resistance training protocol with either 1 or 3 minutes of rest between sets. Training consisted of 7 exercises for the total body of 8 to 12 reps carried out 3 days per week.	8 weeks	No	Ultrasound (elbow flexors, elbow extensors, quadriceps)	Significantly greater increases in anterior thigh muscle thickness and a trend for greater increases in the triceps brachii thickness for the longer rest interval condition
Villanueva et al. (260)	22 untrained elderly men	Random assignment to a resistance training protocol with either 1 or 4 minutes of rest between sets. Training consisted of 2 or 3 sets of 4 to 6 reps carried out 3 days per week.	8 weeks	No	DXA	Significantly greater increases in lean body mass for the shorter rest interval condition

Abbreviations: RM = repetition maximum; CSA = cross-sectional area; MRI = magnetic resonance imaging; DXA = dual X-ray absorptiometry; BIA = bioelectrical impedance analysis; FFM = fat-free mass.

PRACTICAL APPLICATIONS

REST INTERVAL LENGTH

Despite a theoretical concept that shorter rest intervals produce superior muscular adaptations, current research does not support such a contention. In fact, longer intersets rest periods may enhance hypertrophy by allowing for maintenance of a greater volume load. Thus, resistance training protocols should generally provide rest periods of at least 2 minutes to maximize the hypertrophic response, at least when performing multi-joint free weight exercises.

That said, it may be beneficial to employ rest intervals of approximately 60 to 90 seconds for single-joint and perhaps certain machine-based exercises because these movements do not show a reduction in volume load from shorter rest, and the heightened metabolic stress may perhaps confer additional anabolic advantages. Evidence suggests that consistently training with shorter rest intervals promotes adaptations that facilitate the ability to sustain a significantly higher mean percentage of 1RM during training (124). These adaptations include increased capillary and mitochondrial density as well as an improved capacity to buffer hydrogen ions and shuttle them out of muscle, thereby minimizing performance decrements. Conceivably, this could allow maintenance of volume with even greater levels of metabolic stress, potentially enhancing anabolism.

Repetition Duration

Repetition duration represents the sum of the concentric, eccentric, and isometric components of a repetition, and is predicated on the tempo at which the repetition is performed (173). *Tempo* is often expressed as a three-digit arrangement in which the first number is the time (in seconds) to complete the concentric action, the second number is the isometric transition phase between concentric and eccentric actions, and the third number is the time to complete the eccentric action (173). For example, a tempo of 2-0-3 would indicate a repetition taking 2 seconds on the concentric action, not pausing at the top of the movement, and then taking 3 seconds to perform the eccentric action. In the preceding example, the repetition duration would be 5 seconds.

To a certain degree, tempo can be volitionally manipulated. The extent depends on two factors: the intensity of load and the accumulated fatigue. Heavier loads take longer to lift; the closer the load is to the person's 1RM, the slower the concentric action will be, even when the intent is to move the weight as quickly as possible. Moreover, the onset of fatigue causes velocity to decrease because of the inability of working fibers to maintain force output. The capacity to lift even very light loads is curtailed when repetitions approach failure. In one study, the first three concentric repetitions of a 5RM bench press took approximately 1.2 to 1.6 seconds to complete, whereas the fourth and fifth repetitions took 2.5 to 3.3 seconds, respectively (155). These results were seen despite the fact that subjects attempted to lift explosively on all repetitions.

The use of loads of $\leq 80\%$ of 1RM allows lifters to vary concentric lifting cadence; lighter loads enhance this ability. Given that eccentric strength is approximately 20% to 50% greater than concentric strength (12), the velocity of eccentric actions can be altered at loads in excess of concentric 1RM. Some have speculated that intentionally extending the duration of repetitions leads to a superior hypertrophic response as a result of the longer time under load (118).

A systematic review and meta-analysis from our group examined whether alterations in repetition duration affect the hypertrophic response to resistance training (209). Studies met inclusion criteria if they were randomized trials that directly

compared training tempos in dynamic exercise using both concentric and eccentric repetitions carried out to momentary muscular failure. Eight studies ultimately met inclusion criteria, comprising a total of 204 subjects. Repetition duration was stratified into four groups: fast/heavy (sets of 6 to 12 repetitions with a total repetition duration of 0.5 to 4 seconds), fast/light (sets of 20 to 30 with a total repetition duration of 0.5 to 4 seconds), medium (sets of 6 to 12 with a total repetition duration of 4 to 8 seconds), or light (sets of 6 to 12 with a total repetition duration of >8 seconds). Results of the meta-analysis showed no significant differences in muscle hypertrophy in the training durations evaluated. When considering just the studies that employed traditional dynamic constant external resistance (i.e., isotonic) training, it can be inferred that there are no discernable differences in hypertrophy using durations up to approximately 6 seconds.

Subanalysis of data indicates that superslow training is likely detrimental to maximizing hypertrophy. Keogh and colleagues (118) assessed muscle activation in a group of trained lifters during the bench press under a variety of training conditions, including a very slow tempo and a traditional tempo. Those in the slow lifting condition used a repetition duration of 10 seconds (5 seconds for both concentric and eccentric actions), whereas those in the traditional training condition attempted to lift the load as fast as possible. Each condition was carried out to the point of concentric muscular failure. In comparison to the slow tempo, mean EMG activity of the pectoralis major during traditional lifting was markedly higher on the concentric portion of the movement (by 18%, 19%, and 12% for the first, middle, and last repetition, respectively). During eccentric actions, the activation advantage for training at a traditional versus a slow tempo increased to 32%, 36%, and 36% in the first, middle, and last repetition, respectively. These findings provide evidence that volitionally slowing the tempo during a repetition is suboptimal for maximally activating the target muscle.

In the only study to date that employed site-specific measures to evaluate muscle hypertrophy subsequent to superslow versus traditional training, Schuenke and colleagues (221) randomized untrained young females to perform multiple sets of the squat, leg press, and knee extension 2 or

3 days a week for 6 weeks. The superslow group carried out repetitions using a 14-second duration (10 seconds concentric, 4 seconds eccentric); the traditional training group employed a tempo of 1 to 2 seconds on both concentric and eccentric actions. Both groups performed 6RM to 10RM per set, but the loading when training in superslow fashion was much lighter than when using a traditional tempo (approximately 40% to 60% of 1RM vs. approximately 80% to 85% of 1RM, respectively) to allow maintenance of the target repetition range. Post-study increases in Type IIa and Type IIx fibers were substantially greater using a traditional tempo (approximately 33% and 37%, respectively) versus superslow training (approximately 12% and 19%, respectively). In addition, there was a greater decrease in total Type IIx fiber area in the traditional group compared to the superslow group (approximately 39% vs. 28%, respectively), along with a correspondingly greater increase in total Type IIa fiber area (approximately 30% vs. 11%, respectively). This implies that lifting at a volitionally very slow cadence does not stimulate the highest-threshold motor units. Follow-up work from the same lab found that satellite cell content was significantly greater after traditional compared to superslow training across fiber types as well (100).

With respect to the individual muscle actions, some investigators have postulated that intentionally slowing concentric velocity reduces the momentum during a repetition, thereby heightening the tension on a muscle (270). Hypothetically, increased mechanical tension could positively mediate intracellular anabolic signaling, promoting a greater hypertrophic response. It has been shown, however, that the effects of momentum are inconsequential in a concentric movement of 2 seconds versus 10 seconds when the load is kept constant (114). A potential downside of lifting very quickly is a reduction in metabolic stress. Performing the concentric phase of a repetition at 2 seconds resulted in a greater lactate accumulation compared to an explosive concentric contraction despite an equated volume and lower power in the slower cadence (eccentric repetitions were standardized at 2 seconds) (145). The residual effects of this observation on hypertrophy are not clear.

Nogueira and colleagues (169) found that performing concentric actions explosively with

KEY POINT

Current evidence suggests that little difference exists in muscle hypertrophy when training at isotonic repetition durations from 0.5 to 6 seconds. Training at very slow volitional durations (>10 seconds per repetition) appears to produce inferior increases in muscle growth.

a 1-second concentric repetition produced greater increases in muscle thickness compared to performing the repetitions at 2 to 3 seconds. A limitation of the study was that both groups used light loads (40% to 60% of 1RM), and sets were terminated well short of muscular failure. Thus, the design would have provided a bias to the 1-second condition because faster velocities promote greater recruitment and stimulation of higher-threshold motor units in the absence of fatigue (237).

Some have theorized that performing eccentric actions at higher velocities enhances anabolism as a result of increased tension on muscle during high-speed lengthening. Roschel and colleagues (198) found similar activation of Akt, mTOR, and p70^{S6K} following 5 sets of 8 eccentric repetitions at a slow (20° per second) versus fast (210° per second) velocity, suggesting that the velocity of eccentric actions does not influence intracellular anabolic signaling. Several studies have shown a benefit from faster eccentric actions. Shepstone and colleagues (228) reported a trend for greater increases in muscle cross-sectional area of the elbow flexors with faster eccentric repetitions (210° per second vs. 20° per second) and Farthing and Chilibeck (64) demonstrated that fast (180° per second) eccentric actions produced greater increases in muscle thickness as compared to both slow (30° per second) and fast concentric actions, but not slow eccentric actions. It should be noted that all of these studies used isokinetic dynamometry, and the results therefore cannot necessarily be generalized to traditional isotonic training methods using coupled concentric and eccentric actions.

There is evidence that the eccentric tempo may have an impact on hypertrophic results during traditional isotonic training. Assis-Pereira and colleagues (9) reported greater increases in

muscle thickness of the elbow flexors when using an eccentric tempo of 4 seconds versus 1 second during biceps curls (6.3% vs. 16.6%, respectively); the concentric action was performed at a tempo of 1 second in both groups. Furthering this line of research, Shibata and colleagues (229) showed similar hypertrophy of the thigh musculature with eccentric tempos of 4 versus 2 seconds in the squat, with both groups taking 2 seconds to perform the concentric actions. An unpublished study from our group lends support to these findings, with similar increases in muscle thickness noted when training at a 1-0-2 versus 1-0-4 tempo. Although mechanisms cannot be discerned, it is reasonable to speculate that differences between study results are due to the level of control exerted during the respective eccentric actions. A relatively fast eccentric tempo (i.e., 1 second) would seemingly allow gravity to take over a majority of the work, with limited muscular involvement in lowering the load. On the other hand, slowing the eccentric tempo so that the working muscles are forced to exert a braking action provides sufficient mechanical tension to initiate an anabolic response. Based on the evidence, it appears that a 2-second eccentric action is adequate for ensuring complete muscular stimulation during the lengthening component of a repetition; longer durations do not seem to confer additional benefit.

Some evidence suggests that the isometric component at the bottom phase of movement should be minimized to maintain constant tension on the target muscle. Tanimoto and Ishii (244) found that untrained young men performing 12 weeks of knee extensions using a 3-second concentric/eccentric cadence with no rest between eccentric and concentric repetitions experienced a similar hypertrophic response as subjects using a 1-second concentric/eccentric cadence while relaxing for 1 second after each eccentric action. These results were seen despite the use of substantially heavier loads in the faster versus slower cadence conditions (~80% vs. ~50% of 1RM, respectively). On the surface, it is tempting to speculate that the lack of a relaxation phase in the slow cadence condition positively mediated results, perhaps via effects associated with increased ischemia and hypoxia. However, the fact that other aspects of the study were not

controlled (i.e., concentric and eccentric tempo, intensity of load) clouds the ability to draw firm conclusions on the topic.

Attentional focus is perhaps the most important consideration in regard to repetition duration. Simply stated, attentional focus refers to what a person thinks about when carrying out a given motor task. Numerous EMG studies show that greater muscle activation can be achieved by developing a mind-muscle connection (i.e., internal focus of attention) in which the target muscle is actively visualized and consciously forced to contract during exercise performance (36, 37, 234). A recent study from my lab indicates that these findings may extend to longitudinal muscle growth (217). A cohort of untrained young men were randomized to perform 4 sets of 8RM to 12RM of the leg extension and biceps curl using either an internal focus (subjects were repeatedly encouraged to squeeze the muscle on each rep) or an external focus (subjects were repeatedly instructed to get the weight up). After 8 weeks, the group employing the internal focus showed significantly greater increases in elbow flexor muscle thickness compared to the external-focus group (12.4% vs. 6.9%, respectively); in contrast, similar hypertrophic changes were observed between conditions for the quadriceps. We speculated that the differences between muscles may be a function of their use in everyday life. Namely, the arms are frequently used to perform fine motor skills such as lifting delicate objects, and thus the connection between the mind and upper-extremity musculature tends to be stronger to ensure these tasks are properly executed. Alternatively, the legs are most often used for gross motor tasks such as ambulation; hence, the connection between the mind and the lower-extremity musculature tends to be weaker because the associated tasks do not require high levels of concentration. Although more research is needed on the topic, the findings suggest that developing a mind-muscle connection may have greater relevance to hypertrophy than training at a specific tempo; provided loads are lifted with a conscious effort to make the muscle do the work, the tempo is basically moot.

Table 4.7 provides a summary of the research related to repetition duration and muscle hypertrophy.

PRACTICAL APPLICATIONS**REPETITION DURATION**

Current evidence suggests little difference in muscle hypertrophy when training with isotonic repetition durations ranging from 0.5 to 6 seconds to muscular failure. Thus, it would seem that a fairly wide range of repetition durations can be used if the primary goal is to maximize muscle growth. Research is limited on the topic, making it difficult to draw concrete conclusions. Concentric tempos of 1 to 3 seconds can be considered viable options; an eccentric tempo of at least 2 seconds appears necessary to ensure loads are lowered under muscular control. On the other hand, training at very slow volitional durations (>10 seconds per repetition) appears to produce inferior increases in muscle growth, although a lack of controlled studies on the topic makes it difficult to draw definitive conclusions. It is conceivable that combining different repetition durations could enhance the hypertrophic response to resistance training, although this hypothesis requires further study.

Developing a strong mind–muscle connection is perhaps the most important consideration in regard to repetition duration. By focusing on actively contracting the target muscle throughout the range of motion of a given exercise, maximal mechanical forces are directed to the musculature, heightening the degree of stimulation.

TABLE 4.7 Summary of Hypertrophy Training Studies Investigating Repetition Duration

Study	Subjects	Design	Repetition duration	Study duration	Load matched?	Hypertrophy measurement	Findings
Carlson et al. (43)	59 trained young men and women	Random assignment to a single-set, total-body resistance training protocol using a moderate, slow, or very slow cadence carried out twice per week.	6 seconds vs. 20 seconds vs. 90 seconds	10 weeks	Yes	BodPod	No significant differences in lean mass between conditions. Relative increases in lean mass favored the faster-tempo condition.
Clafflin et al. (49)	63 untrained young and old men and women	Random assignment to a resistance training protocol at either a high velocity (hip 250° to 350° per second, knee 100° to 160° per second) or low velocity (hip 30° to 90° per second, knee 20° to 40° per second). All subjects performed 2 sets of 10 reps with a third set that induced failure using 5 to 15 reps. Training was carried out 3 days per week.	0.5 to 0.66 seconds vs. 1 to 2 seconds vs. 2 to 6 seconds vs. 4 to 8 seconds	14 weeks	No	Muscle biopsy (quadriceps)	No effect of training on Type I fibers; 8.2% increase in Type II fibers irrespective of tempo

Study	Subjects	Design	Repetition duration	Study duration	Load matched?	Hypertrophy measurement	Findings
Keeler et al. (117)	14 untrained young and middle-aged women	Random assignment to either superslow or traditional Nautilus resistance training protocol. Subjects performed 1 set of 8RM to 12RM for 8 exercises targeting the entire body. Training was carried out 3 days per week.	6 seconds vs. 15 seconds	10 weeks	No	BodPod	No significant differences in body composition
Munn et al. (160)	115 untrained young men and women	Random assignment to a resistance training protocol of 1 or 3 sets of elbow flexion exercise in either a slow or fast fashion. Training was at 6RM to 8RM for 3 days per week.	2 seconds vs. 6 seconds	6 weeks	No	Skinfold and circumference measurements	No significant differences in lean mass between conditions
Neils et al. (165)	16 untrained young men and women	Random assignment to a protocol of either superslow at 50% of 1RM or traditional resistance training at 80% of 1RM. All subjects performed 1 set of 6RM to 8RM for 7 exercises targeting the entire body. Training was carried out 3 days per week.	6 seconds vs. 15 seconds	8 weeks	No	DXA	No significant differences in body composition between conditions
Nogueira et al. (169)	20 untrained elderly men	Random assignment to an equal work output resistance training protocol in which concentric actions were performed either as fast as possible or at a cadence of 2 to 3 seconds. All subjects performed 3 sets of 8 reps of 7 exercises targeting the entire body. Loads were 40% to 60% of 1RM, and eccentric tempo was 2 to 3 seconds for both conditions. Training was carried out twice weekly.	3 to 4 seconds vs. 4 to 6 seconds	10 weeks	No	Ultrasound (elbow flexors, quadriceps)	Significantly greater increases in thickness of the biceps brachii for the fast condition
Rana et al. (188)	34 untrained young women	Random assignment to a resistance training protocol of moderate intensity (80% to 85% of 1RM) at a tempo of 1 to 2 seconds, low intensity (~40% to 60% of 1RM) at a tempo of 1 to 2 seconds, or slow speed (~40% to 60% of 1RM) at a tempo of 10 seconds concentric and 4 seconds eccentric. All subjects performed 3 sets of 6RM to 10RM of 3 lower-body exercises. Training was carried out 2 or 3 days per week.	2 to 4 seconds vs. 14 seconds	6 weeks	No	BodPod	No significant differences in FFM between conditions

(continued)

Table 4.7 (continued)

Study	Subjects	Design	Repetition duration	Study duration	Load matched?	Hypertrophy measurement	Findings
Schuenke et al. (221)	34 untrained young women	Random assignment to a resistance training protocol of moderate intensity (80% to 85% of 1RM) at a tempo of 1 to 2 seconds, low intensity (~40% to 60% of 1RM) at a tempo of 1 to 2 seconds, or slow speed (~40% to 60% of 1RM) at a tempo of 10 seconds concentric and 4 seconds eccentric. All subjects performed 3 sets of 6RM to 10RM of 3 lower-body exercises. Training was carried out 2 or 3 days per week.	2 to 4 seconds vs. 14 seconds	6 weeks	No	Muscle biopsy (quadriceps)	Significantly greater increases in CSA for the faster condition
Tanimoto and Ishii (244)	24 untrained young men	Random assignment to 50% of 1RM with a 6-second tempo and no relaxing phase between reps, ~80% of 1RM with a 2-second tempo and 1 second of relaxation between reps, or ~50% of 1RM with a 2-second tempo and 1 second of relaxation between reps. All subjects performed 3 sets at 8RM (264) of knee extension exercises. Training was carried out 3 days per week.	2 seconds vs. 6 seconds	12 weeks	No	MRI (quadriceps)	No significant differences in muscle CSA between conditions
Tanimoto et al. (245)	36 untrained young men	Random assignment to ~55% to 60% of 1RM with a 6-second tempo and no relaxing phase between reps or ~80% to 90% of 1RM with a 2-second tempo and 1 second of relaxation between reps. All subjects performed 3 sets at 8RM of 5 exercises targeting the entire body. Training was carried out twice weekly.	2 seconds vs. 6 seconds	13 weeks	No	Ultrasound (chest, anterior and posterior upper arm, abdomen, subscapular, anterior and posterior upper thigh)	No significant differences in muscle thickness between conditions

Study	Subjects	Design	Repetition duration	Study duration	Load matched?	Hypertrophy measurement	Findings
Watanabe et al. (264)	40 untrained elderly men and women	Random assignment to a resistance training protocol of a 6-second tempo and no relaxing phase between reps or a 2-second tempo and 1 second of relaxation between reps. All subjects performed 3 sets of 8 reps at 50% of 1RM of knee extension and knee flexion exercises. Training was carried out twice weekly.	2 seconds vs. 6 seconds	10 weeks	No	Ultrasound (quadriceps, hamstrings)	Significantly greater quadriceps thickness for the slow condition
Watanabe et al. (265)	18 untrained elderly men and women	Random assignment to a resistance training protocol of a 6-second tempo and no relaxing phase between reps or a 2-second tempo and 1 second of relaxation between reps. All subjects performed 3 sets of 13 reps at 30% of 1RM of knee extension exercises. Training was carried out twice weekly.	2 seconds vs. 6 seconds	12 weeks	No	MRI (midthigh)	Significantly greater increases in quadriceps hypertrophy for the slow condition
Young and Bilby (274)	18 untrained young men	Random assignment to a resistance training protocol of either fast concentric contractions or slow controlled movements. All subjects performed 4 sets at 8RM to 12RM of the barbell half-squat exercise. Training was carried out 3 days per week.	2 seconds vs. 4 to 6 seconds	7.5 weeks	No	Ultrasound (quadriceps)	No significant differences in muscle thickness between conditions

Abbreviations: RM = repetition maximum; DXA = dual X-ray absorptiometry; MRI = magnetic resonance imaging; CSA = cross-sectional area; FFM = fat-free mass.

Reprinted from B.J. Schoenfeld, D.I. Ogborn, and J.W. Krieger, "Effect of Repetition Duration During Resistance Training on Muscle Hypertrophy: A Systematic Review and Meta-Analysis," *Sports Medicine* 45, no. 4 (2015): 577-585, with kind permission from Springer Science + Business Media.

PRACTICAL APPLICATIONS**IS THERE AN IDEAL TIME UNDER TENSION TO MAXIMIZE MUSCLE GROWTH?**

While resistance training volume is generally thought of in terms of sets, repetitions, and total work, a concept called time under tension (TUT) also can be considered a relevant variable. TUT can be operationally defined as the total amount of time that a muscle, or group of muscles, endures mechanical stress during resistance exercise. Anecdotally, some fitness professionals have put forth the claim that sets should have a TUT of 40 to 60 seconds to optimally build muscle.

Research into the role of TUT in muscle development is limited. In one of the few studies that attempted to directly investigate the topic, Burd and colleagues (32) carried out an acute, within-subject design in which subjects performed a leg extension exercise at 30% of 1RM with a slow tempo (6-0-6) with one leg and trained the other leg at the same intensity of load with a fast tempo (1-0-1). Three sets were performed for each condition with a 2-minute rest interval between sets, resulting in a 6-fold greater TUT in the slow-tempo condition. Post-exercise muscle biopsies showed significantly greater increases in myofibrillar protein synthesis and intracellular anabolic signaling favoring the slow-tempo condition; differences primarily manifested 24 to 30 hours after the training bout. While on the surface these findings seemingly support the importance of TUT as a driver of hypertrophy, conclusions were confounded by the fact that people in the slow-tempo condition performed all sets to volitional failure while the number of repetitions performed for the fast-tempo condition were matched to that of the slow-tempo condition. Thus, rather than providing insights into the hypertrophic effects of TUT, the results reinforce the importance of challenging the muscles with a high level of effort for muscle building.

Studies comparing superslow training to traditional training whereby both conditions are performed to volitional fatigue have not shown a benefit from higher TUTs; in fact, evidence indicates training in a traditional fashion produces superior hypertrophy despite a substantially lower TUT (221). A caveat to these findings is that the higher TUT in the superslow condition was at the expense of a much lower intensity of load. How these variables interact with one another to affect muscle growth is not clear.

Despite the paucity of objective evidence, a logical case can be made that TUT does play a role in hypertrophy. However, it appears the effects are more related to the time a muscle is worked over the duration of a training session than to the TUT for a given set. In support of this hypothesis, my lab showed that performing a powerlifting-style workout consisting of 7 sets of 3RM produced increases in muscle growth similar to those of a bodybuilding-style workout consisting of 3 sets of 10RM (208). Although TUT in the powerlifting-style sets was markedly lower than in the bodybuilding-style sets (~9 seconds vs. ~30 seconds, respectively), the total TUT for the training session was approximately equal because of the greater number of sets performed for the powerlifting-style condition. These findings are in contrast to a follow-up study showing that when the total number of sets were equated, a bodybuilding-style workout (10RM) elicited greater hypertrophic adaptations compared to a powerlifting-style workout (3RM) (213). Here, the TUT was markedly greater both during each set as well as over the course of the training session.

It also can be hypothesized that not all repetitions equally contribute to hypertrophy. For example, the initial repetitions in a set of 25RM are relatively easy to

execute; only when fatigue begins to manifest does the set become challenging. In contrast, the initial repetitions during a 6RM set are substantially more challenging to complete from the outset, and conceivably would promote greater anabolic stimulation. A case therefore can be made that the TUT in the 6RM protocol would have greater hypertrophic relevance than that of the higher-repetition set. Hence, to some extent TUT should be considered in the context of the repetition range in which a set is performed and the corresponding duration of repetitions that are challenging to complete.

Another inherent issue with TUT is that it considers the duration of the repetitions as a whole and thus neglects to take into account the individual portion of the actions. For example, a set carried out at a 4-0-1 tempo (4-second concentric actions, 1-second eccentric actions) would have the same TUT as a set carried out at a 1-0-4 tempo (1-second concentric actions, 4-second eccentric actions), provided the number of repetitions is equated between sets. This has potentially important implications given the research that shows differential intracellular signaling and hypertrophy responses (72) between concentric and eccentric actions.

All things considered, evidence indicates that TUT plays a role in muscle hypertrophy. However, its implications must be considered in the context of the resistance training variables comprising a given routine (i.e., repetition range, tempo of eccentric versus concentric actions). Within limits, it appears that the total TUT accumulated for a muscle group in a given session, or perhaps over time (e.g., weekly), has the most relevance from a muscle growth standpoint. A rationale for speculation exists whereby a longer TUT (>60 seconds per set) may be beneficial for targeting hypertrophy of Type I muscle fibers; this hypothesis warrants further exploration.

Exercise Order

Current resistance training guidelines prescribe placing large-muscle, multi-joint exercises early in a workout, and placing small-muscle, single-joint movements later (5). These recommendations are based on the premise that the performance of multi-joint exercises is impaired when the smaller secondary synergists are prefatigued by prior single-joint exercises. For example, performance of the arm curl conceivably would fatigue the biceps brachii, thereby impeding the ability to overload the larger latissimus dorsi muscle during subsequent performance of the lat pulldown.

Despite wide acceptance that exercise order should proceed from large- to small-muscle groups, research is equivocal on the topic with respect to hypertrophic outcomes. Acute studies show that performance, as determined by the number of repetitions performed, is compromised in exercises performed toward the end of a session regardless of the size of the muscle trained (232). However, given the heavier loads used during multi-joint movements, the absolute magnitude of the decreases are generally greater in these exercises when they are performed after those involving small-muscle groups. Thus, volume load tends to be better preserved when large-muscle exercises are placed early in the training bout.

Several studies have attempted to directly quantify the effects of exercise order on muscle hypertrophy. Simao and colleagues (231) investigated the performance of upper-body exercises when progressing from large- to small-muscle groups compared to small- to large-muscle groups in untrained men. Exercises included the bench press, lat pulldown, triceps extension, and arm curl. Training was carried out twice per week for 12 weeks. Muscle thickness of the triceps brachii increased only in the group that performed small-muscle-group exercises first, although differences in the thickness of the biceps were similar on an absolute basis. The same lab replicated this basic study design and similarly found greater increases in triceps thickness when the order of exercises progressed from small- to large-muscle groups (238). Although these findings might seem to indicate a benefit to performing smaller-muscle-group exercises first, it should be noted that hypertrophy of the larger muscles was not assessed in either study. It is possible, if not likely, that whichever muscles were worked earlier in the session hypertrophied to a greater extent than those performed toward the end of the bout. This suggests a benefit to prioritizing exercise order so that lagging muscles are worked at the onset of a workout.

It has been postulated that lower-body exercise should precede upper-body exercise. This is based on the hypothesis that lower-body exercise causes a hypoperfusion that compromises the delivery of anabolic hormones to

KEY POINT

Despite widespread belief that exercise order should proceed from large- to small-muscle groups, the hypertrophic benefit has not been demonstrated in controlled research studies.

the upper-body musculature when performed after arm training (269). Ronnestad and colleagues (197) found that hypertrophy of the elbow flexors was magnified when training these muscles was preceded by lower-body exercise, ostensibly as a result of an increase in post-exercise hormonal elevations. These results are in contrast to those of West and colleagues (268), who showed that performing lower-body exercise after arm training did not amplify elbow flexor hypertrophy. The disparate findings between these studies call into question whether there is a hypertrophic advantage to performing lower-body exercise before upper-body exercise. Subsequent work by West and colleagues (269) demonstrated that delivery of testosterone, GH, and IGF-1 to the elbow flexors was not influenced by exercise order. Moreover, the impact of acute systemic fluctuations is of questionable significance and likely has, at best, a small impact on the hypertrophic response (see chapter 2).

Table 4.8 provides a summary of the research related to exercise order and muscle hypertrophy.

PRACTICAL APPLICATIONS

EXERCISE ORDER

Evidence indicates a hypertrophic benefit for muscles worked first in a resistance training bout. Therefore, exercise order should be prioritized so that lagging muscles are trained earlier in the session. In this way, the person expends the greatest energy and focus on the sets of most importance. Whether the muscle group is large or small is of secondary concern.

TABLE 4.8 Summary of Hypertrophy Training Studies Investigating Exercise Order

Study	Subjects	Design	Study duration	Hypertrophy measurement	Findings
Avelar et al. (10)	36 untrained young men	Random assignment to a total-body resistance training protocol in which exercises were performed either from multi-joint to single joint or vice versa. The protocol consisted of 3 sets of 8 to 12 reps performed 3 times per week.	6 weeks	DXA, ultrasound (elbow flexors, quadriceps)	No significant differences in lean soft-tissue mass between conditions. Relative increases in lean soft-tissue mass favored the multi-joint to single-joint condition. No significant differences in muscle thickness were observed between conditions. Relative increases in muscle thickness of the quadriceps favored the multi-joint to single-joint condition.
Cardozo et al. (42)	30 untrained older women	Random assignment to a total-body resistance training protocol in which exercises were performed either from multi-joint to single joint or vice versa. Training was carried out twice per week in circuit fashion, with 3 sets of 8 to 10 repetitions performed for each exercise.	12 weeks	Skinfold measurements	No significant differences in lean mass between conditions
Fisher et al. (70)	25 resistance-trained middle-aged men and women	Random assignment to a resistance training protocol in which exercises were performed either from multi-joint to single joint or rotating between a single-joint exercise followed by a multi-joint exercise. All subjects performed a single set at a moderate intensity of load to muscular failure. Training was carried out 2 days per week.	12 weeks	BodPod	No significant differences in lean mass between conditions
Simao et al. (231)	31 recreationally trained young men	Random assignment to a resistance training protocol in which exercise order began either with large- and progressed to small-muscle-group exercises or began with small- and progressed to large-muscle-group exercises. The protocol consisted of 2 to 4 sets of 4 upper-body exercises: 2 compound movements and 2 single-joint movements were carried out twice per week. Intensity of load was periodized from light to heavy each month over the course of the study, descending from 12-15RM to 3-5RM.	12 weeks	Ultrasound (elbow flexors, elbow extensors)	No significant differences in thickness of the biceps or triceps between conditions

(continued)

Table 4.8 (continued)

Study	Subjects	Design	Study duration	Hypertrophy measurement	Findings
Spinetti et al. (238)	30 recreationally trained young men	Random assignment to a resistance training protocol in which exercise order either began with large- and progressed to small-muscle-group exercises or began with small- and progressed to large-muscle-group exercises. The protocol consisted of 2 to 4 sets of 4 upper-body exercises: 2 compound movements and 2 single-joint movements were carried out twice per week. Intensity of load was carried out in an undulating periodized fashion alternating between light (12RM to 15RM), moderate (8RM to 10RM), and heavy (3RM to 5RM).	12 weeks	Ultrasound (elbow flexors, elbow extensors)	No significant differences in thickness of the biceps or triceps between conditions
Tomeleri et al. (255)	29 untrained older women	Random assignment to a total-body resistance training protocol in which exercises were performed either from multi-joint to single joint or vice versa. Both groups performed 3 sets of 10 to 15 reps per exercise carried out 3 days per week	12 weeks	DXA	No significant differences in lean soft-tissue mass between conditions

Abbreviations: RM = repetition maximum; DXA = dual X-ray absorptiometry.

Range of Motion

Basic principles of structural anatomy and kinesiology dictate that muscles have greater contributions at different joint angles for given exercises. For example, there is evidence that the quadriceps muscles are differentially activated during knee extension exercise: The vastus lateralis is maximally activated during the first 60° of range of motion (ROM), whereas the vastus medialis is maximally activated during the final 60° of ROM (230). Similar findings have been reported during the arm curl: The short head appears to be more active in the latter phase of the movement (i.e., greater elbow flexion), whereas the long head is more active in the early phase (28).

When comparing partial and complete ROMs, the body of literature generally shows a hypertrophic benefit to training through a full ROM. This has been displayed in both upper- and lower-body muscles using a variety of exercises. Pinto and colleagues (180) showed that full ROM training of the elbow flexors

(0° to 130° of flexion) produced greater increases in muscle thickness compared to partial-range training (50° to 100° of flexion), with the difference in effect size strongly favoring the full ROM condition (0.52). Similarly, McMahon and colleagues (151) showed that although knee extension at full ROM (0° to 90°) and partial ROM (0° to 50°) both increased quadriceps muscle cross-sectional area, the magnitude of hypertrophy was significantly greater at 75% of femur length in the full-range condition. Interestingly, Bloomquist and colleagues (25) showed that deep squats (0° to 120° of knee flexion) promoted increases in cross-sectional area across the entire frontal thigh musculature, whereas shallow squats (0° to 60° of knee flexion) elicited significant growth only in the two most proximal sites. Furthermore, the overall change in cross-sectional area was greater at all measured sites in the deep-squat group.

Recent research suggests the topic may be more nuanced than previously thought. In an 8-week squat study, Kubo and colleagues (127) reported that training through a full ROM (0° to 140°)

KEY POINT

Muscles are activated differentially throughout the range of motion. Full ROM movements should therefore form the basis of a hypertrophy training program, although including some partial-ROM training may provide additional benefit.

elicited significantly greater increases in muscle volume of the adductors and gluteus maximus compared to a partial ROM (0° to 90°). However, no differences were observed in quadriceps muscle volume between conditions, suggesting that the response to variations in ROM may be muscle specific over a given joint excursion. Other research shows similar quadriceps growth when using either a partial ROM (0° to 60° knee flexion) or a full ROM (0° to 100° knee flexion) during isokinetic knee extension on a dynamometer (257), although findings must be taken with the caveat that this mode of training provides accommodating resistance throughout the entire ROM. In the only study to date that included resistance-trained subjects, Goto and colleagues (84) reported greater triceps brachii hypertrophy pursuant to elbow extension exercise using a partial ROM (elbow range from 45° to 90°) versus a full ROM (from 0° to 120°). Intriguingly, the authors noted a positive correlation between markers of intramuscular hypoxia and the percent increase in muscle cross-sectional area ($r = .70$) during partial-ROM training, raising the possibility that maintaining a constant tension on the working muscle through a limited range may heighten anabolism, perhaps via compression of the surrounding vessels.

At present, no studies have endeavored to investigate the possible benefits of combining

partial- and full-ROM training. Evidence suggests that quadriceps muscle activation varies throughout the ROM during knee extension performance (230); the vastus lateralis shows greatest activity at the midportion of the movement, while the activity of the vastus medialis oblique is greatest approaching lockout. Similarly, the long head of the biceps brachii is dominant in the early phase of elbow extension, while the short head becomes more active during the latter phase (28). Moreover, partial-ROM training affords the ability to employ heavier loading during exercise performance, which may in turn facilitate the use of higher magnitudes of load during full-range movements (142). Thus, incorporating partial-range movements into a hypertrophy-oriented program may help to enhance results.

Evidence suggests that training at longer muscle lengths (i.e., when the muscle is in a stretched position) promotes greater hypertrophic adaptations than training at shorter muscle lengths. McMahon and colleagues (150) compared the hypertrophic response to knee extensions at shortened (0° to 50° of knee flexion) or lengthened (40° to 90° of knee flexion) positions. Results showed significantly greater increases in distal cross-sectional area of the quadriceps (53% vs. 18%) as well as fascicle length (29% vs. 14%) in favor of the long- versus short-length training, respectively. Moreover, IGF-1 levels were significantly greater following long-length training than following short-length training (31% vs. 7%, respectively), suggesting that exercise at long muscle lengths induces greater metabolic and mechanical stress. Other research shows a clear hypertrophic advantage to training at longer muscle lengths during knee extension exercises (170). The combination of findings indicates that stretched muscle is in an optimal position for hypertrophy.

Table 4.9 provides a summary of the research related to ROM and muscle hypertrophy.

PRACTICAL APPLICATIONS**RANGE OF MOTION**

Maximal muscle development requires training through a complete ROM. Thus, full ROM movements should form the basis of a hypertrophy-oriented program. The stretched position appears particularly important in eliciting hypertrophic gains. That said, integrating partial-range movements may help to enhance hypertrophy.

TABLE 4.9 Summary of Hypertrophy Training Studies Investigating Range of Motion

Study	Subjects	Design	Study duration	RM equated?	Hypertrophy measurement	Findings
Bloomquist et al. (25)	24 untrained young men	Random assignment to squat training performed as either a deep squat (0° to 120° of knee flexion) or shallow squat (0° to 60° of knee flexion). All subjects performed 3 to 5 sets of 6 to 10 reps for 3 days per week.	12 weeks	Yes	MRI (front and rear thigh), ultrasound (vastus lateralis), DXA	Significantly greater increases in frontal thigh CSA and greater relative gains in lean mass for the 0° to 120° condition
Goto et al. (84)	44 resistance-trained young men	Random assignment to elbow extension exercise using either partial (elbow range from 45° to 90°) or full (from 0° to 120°) ROM. Training consisted of 3 sets of 8 reps carried out 3 days per week.	8 weeks	Yes	Ultrasound (triceps brachii), circumference measurement (upper arm), CSA calculated as the product of muscle thickness and circumference.	Greater increases in elbow extensor CSA for the partial-ROM condition
Kubo et al. (127)	17 untrained young men	Random assignment to squat training using either a partial (0° to 90°) or full (0° to 140°) ROM. Training consisted of 3 sets of 8 to 10 reps at 60% to 90% of 1RM twice per week.	10 weeks	Yes	MRI (quadriceps, hamstrings, adductors, gluteus maximus)	Significantly greater increases in muscle volume of the adductors and gluteus maximus in the full-ROM condition. No significant differences in quadriceps or hamstrings volume between conditions.
McMahon et al. (151)	26 recreationally active young men and women	Random assignment to lower-body training performed either as a full ROM (0° to 90° of knee flexion) or partial ROM (0° to 50° of knee flexion). All subjects performed 3 sets at 80% of 1RM for 3 days per week.	8 weeks	Yes	Ultrasound (vastus lateralis)	Significantly greater increases in vastus lateralis CSA for the full-ROM condition
Pinto et al. (180)	40 untrained young men	Random assignment to elbow flexion exercises with either a full ROM (0° to 130°) or partial ROM (50° to 100°). All subjects performed 2 to 4 sets of 8RM to 20RM twice per week.	10 weeks	Yes	Ultrasound (elbow flexors)	No significant differences between conditions
Valamatos et al. (257)	11 untrained young men	Within-subject design in which subjects' legs were randomly assigned to isokinetic knee extension exercise using a partial ROM (0° to 60°) or a full ROM (0° to 100°). Training consisted of 2 to 7 sets of 6 to 15 reps carried out 3 days per week.	15 weeks	Yes	MRI (vastus lateralis)	No differences in muscle volume of the vastus lateralis between conditions

Abbreviations: RM = repetition maximum; MRI = magnetic resonance imaging; DXA = dual X-ray absorptiometry; ROM = range of motion; CSA = cross-sectional area.

Intensity of Effort

The effort exerted during resistance training, often referred to as *intensity of effort*, can influence exercise-induced hypertrophy. Intensity of effort is generally gauged by the proximity to muscular failure, which is defined as the point during a set at which muscles can no longer produce the force necessary to concentrically lift a given load (205). Although the merits of training to failure are still a matter of debate, it is commonly believed that the practice is necessary for eliciting a maximal hypertrophic response (31, 272).

The primary rationale for training to failure is to maximize motor unit recruitment (272), which is a requisite for achieving maximal protein accretion across all fiber types. Evidence supporting this position is lacking, however. It has been demonstrated that fatiguing contractions result in a corresponding increase in surface EMG activity, presumably as a result of the increased contribution of high-threshold motor units to maintain force output as lower-threshold motor units fatigue (237). However, as previously mentioned, surface EMG is not specific to recruitment; increases in amplitude can be caused by several other factors as well, including rate coding, synchronization, muscle fiber propagation velocity, and intracellular action potentials (17, 57).

The extent of motor unit activation likely depends on the magnitude of load. During heavy-load training, the highest-threshold motor units are recruited almost immediately, whereas during lighter-load training, the recruitment of these motor units is delayed. The point at which complete motor unit activation occurs is not clear, but evidence suggests that a majority of the motor unit pool for a working muscle is recruited with loads as low as 30% of 1RM, provided sets are carried out with a high intensity of effort (158). Thus, a high intensity of effort becomes increasingly important as the intensity of loading is reduced.

Training to failure may also enhance hypertrophy by increasing metabolic stress. Continuing to train under conditions of anaerobic glycolysis heightens the buildup of metabolites, which theoretically augments post-exercise anabolism. Moreover, the continued compression of vessels induces greater acute hypoxia in the working muscles, which may further contribute to hypertrophic adaptations (222).

Despite the important implications of the topic for muscle development, controlled research into the effects of failure training on hypertrophic adaptations remains somewhat limited. Initial work by Goto and colleagues (83) compared hypertrophic adaptations between two groups of recreationally trained men performing 3 to 5 sets of 10 repetitions with an interset rest period of 60 seconds. One group performed repetitions continuously to failure, and the other group took a 30-second rest period at the midpoint of each set. After 12 weeks, muscle cross-sectional area was markedly greater in the group that carried out training to failure compared to the group that did not. Although these results are intriguing, the style of training does not replicate a traditional nonfailure approach in which sets are stopped just short of all-out effort. At most, the study shows that stopping well short of failure attenuates hypertrophic adaptations.

Studies that have endeavored to study the topic more directly show conflicting results. Giessing and colleagues (79) reported that well-trained subjects gained significantly greater lean mass when training to muscular failure at 80% of 1RM than when using a self-determined ter-

KEY POINT

Evidence that training to failure maximizes motor unit recruitment is lacking, although other benefits of training to failure have been shown.

mination of a set at 60% of 1RM. Limitations of the study include the use of a single-set training protocol, which as previously discussed is suboptimal for maximal hypertrophic gains, and different intensities of load between conditions. Results of a study by Martorelli and colleagues (140) lend some support to these findings, showing markedly greater increases in biceps brachii thickness in a cohort of active women who performed bilateral arm curls to failure compared to those who stopped short of failure (17.5% vs. 8.5%, respectively).

Conversely, Sampson and Groeller (200) found no differences between training to failure at 85% of 1RM and stopping 2 repetitions short of failure at this intensity of load in a cohort of untrained men. The study was confounded by the fact that the nonfailure group performed a single set to failure at the end of each week to determine loading for the subsequent week. It is not clear whether this factor influenced results. Findings are consistent with those of Nobrega and colleagues (168) who showed similar increases in quadriceps cross-sectional area when training with both high (80% of 1RM) and low (30% of 1RM) loads either to failure or terminating sets at the point at which participants voluntarily decided to stop. However, no specific instructions were provided to avoid reaching failure and, given that volume loads were similar across conditions, it can be inferred that sets in the nonfailure condition were performed at close to full fatigue. Research in community-dwelling older men (~66 years of age) also indicates no hypertrophic benefit to failure training versus performing sets at 50% of 1RM and doubling the number of sets to equate volume load (53). However, a group performing the same nonfailure protocol with an equal number of sets as those training to failure showed only minimal muscle growth across the study period. Alternatively, a similar study of resistance-trained men using either volume-equated sets to failure (4 sets of 10 repetitions at 10RM per exercise with a 2-minute rest) or sets not performed to failure (8 sets of 5 repetitions at 10RM per exercise with 1-minute rest) showed greater hypertrophic increases in the group training to failure (116). These findings indicate that volume is a more important hypertrophic variable than intensity of effort

in untrained older men, but proximity to failure becomes increasingly more important in younger individuals with experience in resistance training.

Recently, Carroll and colleagues (46) randomized well-trained men to perform a volume load-equated total-body resistance training routine with sets either taken to failure or guided by submaximal percentages of 1RM, in which failure was not reached in any set. After 10 weeks, the group that used submaximal 1RM percentages to guide training achieved greater increases in cross-sectional area of the vastus lateralis and greater increases in individual Type I and Type II fiber cross-sectional area than the group training to failure. Although these findings suggest that managing volume by systematically stopping short of failure may enhance hypertrophic results, the results should be interpreted with the caveat that much of the training was carried out using very heavy loads (≤ 5 RM), thus limiting extrapolation to the higher-repetition work more often used in bodybuilding routines.

A potential issue with failure training is that it increases the potential for overtraining and psychological burnout when carried out regularly over time (74). Izquierdo and colleagues (110) reported reductions in resting IGF-1 concentrations and a blunting of resting testosterone levels in a group of physically active men when failure training was consistently employed over the course of a 16-week resistance training protocol. Such hormonal alterations are consistent with chronic overtraining, suggesting a detrimental effect of repeatedly working to the point of failure. Thus, managing the amount of failure training, if it is to be undertaken, is important to ensuring progression over time.

As with most studies on resistance training variables, the current research compares a group completing sets to failure with a group taking no sets to failure. However, the choice of whether to train to failure does not have to be dichotomous. Evidence suggests that stopping a set a couple of repetitions short of failure does not compromise muscle gains, at least when using moderately heavy loads (6RM to 12RM) on a volume-equated basis. However, a case can be made that selective inclusion of

some failure training may augment muscle development. This is especially important as one becomes more experienced with resistance training, which in turn necessitates progressive challenges to the neuromuscular system to elicit continued hypertrophic adaptations.

The repetitions in reserve (RIR) scale represents a viable strategy to help manage the extent of training to failure (277). In this scale, an RIR of 0 equates to training to failure, an RIR of 1 equates to stopping a set 1 repetition short of failure, an RIR of 2 equates to stopping a set 2 repetitions short of failure, and so on. The scale requires experimentation, but after a period of familiarization, most trained lifters are able to use it to estimate proximity to failure with good precision. When using this scale, most sets should be performed at an RIR of 1 or 2. Failure training could then be selectively employed on the last set of an exercise. This approach not only helps to prevent overtaxing the neuromuscular system, but also in preserving volume load across sets; training to failure on the first set will tend to reduce the number of repetitions achieved on the ensuing sets at a given magnitude of load.

The type of exercise also should be taken into consideration during failure training. Multi-joint exercises such as squats, presses, and rows are highly taxing, both from a central and peripheral standpoint. Limiting the use of failure training in these movements can help to mitigate systemic fatigue and thus diminish the potential for overtraining. On the other hand, failure training can be implemented more freely during performance of single-joint exercises because they are less physically and mentally demanding, and thus their impact on post-exercise recovery is minimized.

Finally, periodization of failure training is a viable way to elicit a supercompensatory response. For example, a higher number of sets are taken to failure during a brief peaking cycle, while fewer failure sets are performed during the other training cycles. This balances training that challenges the neuromuscular system in a manner that spurs hypertrophic adaptation with the necessary recovery to facilitate rejuvenation.

Table 4.10 provides a summary of the research related to intensity of effort and muscle hypertrophy.

PRACTICAL APPLICATIONS

INTENSITY OF EFFORT

Although research remains somewhat equivocal, there is logical rationale for performing at least some sets to failure in a hypertrophy-oriented program, especially in individuals with considerable training experience. This seems to be of particular importance when employing high-repetition training because of the relationship between the proximity to failure and muscle activation during light-load training, and with increasingly greater resistance training experience. However, persistently training to failure increases the potential for nonfunctional overreaching and perhaps overtraining.

When taking all factors into account, it is recommended that most sets are carried out with an RIR of 1 or 2. Failure training should then be implemented selectively, usually reserved for the last set of a given exercise. As a general rule, failure should be used more judiciously with multi-joint exercises, while a more liberal approach can be employed with single-joint movements. The frequency of failure training also can be periodized to bring about a supercompensatory response. An example would be performing an initial cycle in which all sets are stopped a repetition or two short of failure, followed by taking the last set of each exercise to failure, and then culminating in a brief cycle in which the majority of sets are carried out to failure.

TABLE 4.10 Summary of Hypertrophy Training Studies Investigating Intensity of Effort

Study	Subjects	Design	Study duration	Hypertrophy measurement	Findings
Carroll et al. (46)	15 trained young men	Random assignment to a total-body resistance training protocol involving either training to failure or at a given relative intensity based on a percentage of target RM. Training was carried out across a range of loading zones with 3 to 5 sets performed per exercise over 3 days per week.	10 weeks	Muscle biopsy, ultrasound (quadriceps)	Greater increases in muscle thickness, CSA, and fiber type-specific hypertrophy in the nonfailure condition
da Silva et al. (53)	52 untrained older men	Random assignment to one of three groups: (1) a group that performed 2 to 3 sets of reps to failure at 65% to 75% of 1RM, (2) a nonfailure group that performed the same routine as the failure training group, but sets were carried out at 50% of the repetition load, or (3) a group that performed the same routine as the nonfailure group, but with volume equalized to the failure condition by including additional nonfailure sets. Training consisted of the leg press and knee extension performed twice per week.	12 weeks	Ultrasound (quadriceps)	Greater increases in quadriceps muscle thickness for the failure and volume-euated nonfailure conditions vs. the nonfailure condition with unequated volume
Giessing et al. (79)	79 resistance-trained men and women	Random assignment to a resistance training protocol involving either training to self-selected RM at 60% of 1RM or training to momentary muscular failure at 80% of 1RM. All subjects performed a single set of 8 exercises for the entire body. Training was carried out 3 days per week.	10 weeks	BIA	Greater increases in lean mass for the training-to-fatigue condition
Goto et al. (83)	26 untrained young men	Random assignment to perform either 3 to 5 sets of 10RM or the same routine with a 30-second rest at the midpoint of each set so that failure was not induced. All groups performed 2 upper-body exercises and 1 lower-body exercise twice per week.	12 weeks	MRI (quadriceps)	Significantly greater increases in quadriceps CSA for the training-to-fatigue condition
Karsten et al. (116)	18 trained young men	Random assignment to 1 of 2 volume-euated resistance training protocols involving either 4 sets of 10 reps per exercise with 2-minute rest performed to failure or 8 sets of 5 reps per exercise with 1-minute rest not performed to failure.	6 weeks	BodPod, ultrasound (elbow flexors, anterior deltoid, vastus medialis)	Increases in FFM and vastus medialis muscle thickness favored the failure condition. No significant between-group differences in muscle thickness were noted for the anterior deltoid or elbow flexors.
Martorelli et al. (140)	89 active young women	Random assignment to one of three groups: (1) a group that performed 3 sets of reps to failure at 70% of 1RM, (2) a group that performed 4 sets of 7 reps not to failure, but with volume equalized to the failure condition, and (3) a group that performed 3 sets of 7 reps not to failure. Training consisted of free weight biceps curls performed twice per week.	10 weeks	Ultrasound (elbow flexors)	No significant differences in elbow flexor muscle thickness between conditions. Relative increases substantially favored the group training to muscle failure.

Study	Subjects	Design	Study duration	Hypertrophy measurement	Findings
Nobrega et al. (168)	32 untrained young men	Within-subject design in which each lower limb was randomized to perform 3 sets of leg extension exercise twice per week at either a high load (80% of 1RM) or low load (30% of 1RM) with training carried out either to failure or terminated at the point at which participants voluntarily interrupted training.	12 weeks	Ultrasound (quadriceps)	No significant differences in vastus lateralis CSA between conditions
Pareja-Blanco et al. (175)	24 trained young men	Random assignment to perform multiset squat training with sets terminated at either a 20% or 40% velocity loss. Training was carried out twice per week using a load corresponding to ~70% to 85% of 1RM.	8 weeks	MRI, muscle biopsy (quadriceps)	No significant differences in fiber CSA between conditions. Greater increases in whole-muscle vastus lateralis and vastus intermedius CSA in the group that trained closer to failure (40% velocity loss).
Sampson and Groeller (200)	28 untrained young men	Random assignment to perform resisted elbow flexion under one of three conditions: (1) a control condition that performed both concentric and eccentric components at a speed of 2 seconds, (2) a rapid shortening condition that performed maximal acceleration during the concentric action followed by a 2-second eccentric action, or (3) a stretch-shortening group that performed both eccentric and concentric components with maximal acceleration. The control group trained to failure; the other two groups did not. Training consisted of 4 sets at 85% of 1RM performed 3 days per week.	12 weeks	MRI (elbow flexors)	No significant differences in elbow flexor CSA between groups
Schott et al. (220)	7 untrained young men and women	Within-subject design in which subjects performed either an intermittent isometric knee extension protocol consisting of 4 sets of 10 reps lasting 3 seconds with a 2-second rest between reps and a 2-minute rest between sets or a continuous protocol of 4 sets of isometric actions lasting 30 seconds with a 1-minute rest between sets. Training was carried out 3 days per week.	14 weeks	CT (quadriceps)	Greater increases in quadriceps hypertrophy for the training-to-fatigue condition
Teodoro et al. (249)	36 untrained older men	Random assignment to perform the leg extension and leg press under one of three conditions: (1) to failure, (2) not to failure with only 50% of the reps as the failure training group, or (3) not to failure with equal training volume of the failure training group. Training was carried out twice per week with loads ranging from 65% to 80% of 1RM. Additional moderate intensity treadmill aerobic exercise was performed concurrently with the resistance training program.	20 weeks	Ultrasound (quadriceps)	No significant differences in muscle thickness between conditions

Abbreviations: BIA = bioelectrical impedance analysis; RM = repetition maximum; CSA = cross-sectional area; CT = computerized tomography; MRI = magnetic resonance imaging; FFM = fat-free mass.

TAKE-HOME POINTS

- Multiset protocols favoring higher volumes of resistance training optimize the hypertrophic response. A range of 10 to 20 sets per muscle is a general guideline for weekly volume prescription. That said, there is a fairly wide interindividual response to the volume dose, and thus some people will thrive on somewhat lower volumes, while others will benefit from slightly higher volumes. Strategic use of very high volumes (~30+ sets per muscle) can be employed to help bring up lagging muscle groups. To avoid overtraining, overall volume should be progressively increased over the course of a training cycle; periods of reduced training volume should be integrated regularly to facilitate the recovery process.
- When employing lower total volumes, training frequency does not seem to play much if any role in muscle growth. In these cases, individuals can choose the frequency that best fits their schedule and goals. Alternatively, when moderate to higher volumes are performed (>10 sets per muscle per week), higher training frequencies (at least twice per week) allow for better volume management, thus facilitating greater muscular adaptations. Although both total-body and split routines can be viable training strategies, splitting workouts by body region or function (e.g., upper and lower, pushing and pulling) may be superior when training with higher volumes because it allows for higher weekly frequencies (and thus shorter sessions) while affording greater muscular recuperation between workouts.
- Training across a wide spectrum of repetition ranges (1 to 20+) is recommended to ensure complete whole-muscle development. From an efficiency standpoint, there is merit to focusing on a medium-repetition range (6RM to 12RM) and devoting specific training cycles or sessions to lower- and higher-repetition training.
- Once competency in the basic movement patterns has been established, a variety of exercises should be employed over the course of a periodized training program to maximize whole-body muscle hypertrophy, with a particular focus on working muscles based on their anatomical design. This should include the liberal use of free-form (i.e., free weights and cables) and machine-based exercises. Similarly, both multi- and single-joint exercises should be included in a hypertrophy-specific routine to maximize muscular growth.
- Both concentric and eccentric actions should be incorporated during training. Evidence of the benefits of combining isometric actions with dynamic actions is lacking at this time. The addition of supramaximal eccentric loading may enhance the hypertrophic response.
- An optimal rest interval for hypertrophy training does not appear to exist. Research indicates that resting at least 2 minutes between sets provides a hypertrophic advantage over resting for shorter periods, at least when performing multi-joint free weight exercises. It may be beneficial to employ rest intervals of approximately 60 to 90 seconds for single-joint and perhaps certain machine-based exercises because these movements do not show a reduction in volume load from shorter rest, and the heightened metabolic stress may confer additional anabolic advantages.
- Current evidence suggests little difference in muscle hypertrophy when training with isotonic repetition durations ranging from 0.5 to 6 seconds to muscular failure. Thus, a fairly wide range of repetition durations can be employed if the

primary goal is to maximize muscle growth. Training at very slow volitional durations (>10 seconds per repetition) appears to be suboptimal for increasing muscle size and thus should be avoided. A more important consideration is to develop a strong mind–muscle connection, which involves focusing on actively contracting the target muscle throughout the range of motion of a given exercise. If the target muscle is forced to work over the entire concentric and eccentric portions of movement, tempo becomes largely moot.

- Evidence indicates a hypertrophic benefit for muscles worked first in a resistance training bout. Therefore, lagging muscles should be trained earlier in the session.
 - Full-ROM movements should form the basis of a hypertrophy-oriented program. Integrating partial-range movements may enhance hypertrophic adaptations.
 - Hypertrophy-oriented programs should include sets taken to muscular failure as well as those that are terminated short of an all-out effort. As a general rule, most sets should be carried out with an RIR of 1 or 2. Failure training should then be implemented selectively, generally reserved for the last set of a given exercise. Failure training should be used more judiciously with multi-joint exercises, while a more liberal approach can be employed with single-joint movements.
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